4

Stochastic Differential Equations

1. INTRODUCTION

Continuous dynamical systems with a finite-dimensional state, which are subject to random disturbances, can often be represented by a nonlinear ordinary differential equation. Let x_t and w_t be, respectively, the n-dimensional vector state and the m-vector random disturbance at time t. Then a general differential equation of the type described can be written as

$$dx_t/dt = f(x_t, w_t, t), \qquad t \geqslant t_0, \tag{4.1}$$

where f is a nonlinear, real n-vector function. Equation (4.1) is the continuous-time analog of the stochastic difference equation that we studied in Section 3.9. It is called a *stochastic differential equation*, or a *continuous stochastic dynamical system*. The random disturbing function w_t is called a *random forcing* (*driving*, *input*) *function*. The initial condition for (4.1) can be a fixed constant or a random variable x_{t_0} with a specified distribution. The probability law of the $\{w_t, t \geqslant t_0\}$ process is assumed specified.

Now if the function f and the $\{w_t\}$ process are suitably restricted so that the integral

$$\int_{t_0}^{t} f(x_\tau, w_\tau, \tau)\, d\tau$$

is well defined, say in the mean square sense (Section 3.4), and the derivative dx_t/dt is also understood in the mean square sense, then by Theorem 3.10

$$x_t - x_{t_0} = \int_{t_0}^{t} f(x_\tau , w_\tau , \tau)\, d\tau. \tag{4.2}$$

By Theorem 3.9, (4.2) implies (4.1). Thus (4.1) and (4.2) are equivalent. We might now inquire as to the existence, uniqueness, and other properties of the stochastic process generated by (4.2). As in the case of stochastic difference equations, we are interested in the density function $p(x_t)$, $t \geqslant t_0$.

An important special case of (4.1) is the stochastic differential equation with an *additive* white Gaussian forcing function

$$dx_t/dt = f(x_t , t) + G(x_t , t)w_t , \qquad t \geqslant t_0 , \tag{4.3}$$

where G is an $n \times m$ matrix function, and with initial condition x_{t_0} that is independent of the white Gaussian process $\{w_t , t \geqslant t_0\}$. Equation (4.3) is sometimes called a *Langevin* equation. The importance of the white noise model for random disturbances in dynamical systems has already been noted (Section 3.8).

Now the $\{w_t\}$ process is delta-correlated, and therefore, in view of Theorem 3.7, w_t is not mean square Riemann integrable. Its sample functions are delta functions (Example 3.20), and as a result, w_t is not integrable wp 1. Consequently, (4.3) has no mathematical meaning. Recall from Example 3.21 that white Gaussian noise is the *formal* derivative of Brownian motion. Let $\{\beta_t , t \geqslant t_0\}$ be a vector process of independent Brownian motions. Then, formally,

$$w_t \sim d\beta_t/dt,$$

and (4.3) can be considered to be *formally* equivalent to

$$dx_t = f(x_t , t)\, dt + G(x_t , t)\, d\beta_t , \qquad t \geqslant t_0 . \tag{4.4}$$

Now (4.4) is meaningful only insofar as its integral is defined:

$$x_t - x_{t_0} = \int_{t_0}^{t} f(x_\tau , \tau)\, d\tau + \int_{t_0}^{t} G(x_\tau , \tau)\, d\beta_\tau . \tag{4.5}$$

The first integral can be defined as an ms Riemann integral (Section 3.4), or as an ordinary integral for the sample functions. The second integral has not yet been defined. Because of the erratic properties of the

Brownian motion process (unbounded variation; see Section 3.5), it cannot be defined for the sample functions. It has been defined in a mean square sense by Itô [9]. Another definition, also in the mean square sense, has recently been proposed by Stratonovich [17]. We shall be largely concerned here with the second integral in (4.5) as defined in the Itô sense. That integral is then called an *Itô stochastic integral,* and (4.4) is an *Itô stochastic differential equation.*

For the moment, consider the differentials in (4.4) as small increments and write (4.4) as

$$x_{t+\delta t} - x_t = f(x_t, t)\,\delta t + G(x_t, t)(\beta_{t+\delta t} - \beta_t).$$

Then, given x_t, $x_{t+\delta t}$ depends only on the Brownian motion increment $\beta_{t+\delta t} - \beta_t$. Now Brownian motion increments are independent, and by assumption, $\{d\beta_t, t \geqslant t_0\}$ is independent of x_{t_0}. Therefore, with x_t given, $x_{t+\delta t}$ is independent of $\{x_\tau, \tau \leqslant t\}$. Thus the $\{x_t, t \geqslant t_0\}$ process generated by (4.4) (if it exists) is a *Markov* process.

This chapter is devoted to a study of the stochastic integral, the stochastic differential equation (4.4), and the properties of its solution process. That process is Markov, and we shall derive the laws of evolution for its transition probability densities and for the density $p(x_t)$. We shall also be concerned briefly with the relationship between the solutions of (4.4) and physically realizable processes, that is, with the problem of modeling.

2. MODELING THE PROCESS e^{β_t}

The solution x_t of the stochastic differential equation (4.4) will clearly be a function of the Brownian motion process $\{\beta_t\}$. It is instructive to consider such a process x_t and ask what differential equation that process satisfies.

Consider the scalar process defined by

$$x_t = e^{\beta_t}, \qquad t \geqslant 0, \qquad \beta_0 = 0 \qquad \text{wp 1}, \tag{4.6}$$

where $\{\beta_t, t \geqslant 0\}$ is a scalar Brownian motion process with variance parameter σ^2. Now β_t is differentiable nowhere wp 1 and in mean square (Section 3.5). As a result, we cannot differentiate (4.6) to obtain a differential equation for x_t. Consider a power series expansion of (4.6):

$$\delta x_t = e^{\beta_t + \delta\beta_t} - e^{\beta_t} = x_t\left(\delta\beta_t + \frac{\delta\beta_t^2}{2} + \cdots\right), \qquad \delta\beta_t^2 \equiv (\delta\beta_t)^2, \tag{4.7}$$

where $\delta\xi_t = \xi_{t+\delta t} - \xi_t$, $\delta t > 0$. Now if we truncate the expansion at the first-order term, we have

$$\begin{aligned}\mathscr{E}\{\delta x_t - x_t\,\delta\beta_t\} &= \mathscr{E}\left\{x_t\left(\frac{\delta\beta_t^2}{2} + \cdots\right)\right\} = \mathscr{E}\left\{\mathscr{E}\left\{x_t\left(\frac{\delta\beta_t^2}{2} + \cdots\right)\,\middle|\, x_t\right\}\right\} \\ &= \mathscr{E}\left\{x_t\mathscr{E}\left\{\left(\frac{\delta\beta_t^2}{2} + \cdots\right)\,\middle|\, x_t\right\}\right\} = \mathscr{E}\left\{x_t\mathscr{E}\left\{\left(\frac{\delta\beta_t^2}{2} + \cdots\right)\right\}\right\} \\ &= \mathscr{E}\{x_t\}\,\mathscr{E}\left\{\left(\frac{\delta\beta_t^2}{2} + \cdots\right)\right\} = O(\delta t),^1 \end{aligned} \tag{4.8}$$

and, similarly,

$$\mathscr{E}\{|\,\delta x_t - x_t\,\delta\beta_t\,|^2\} = O(\delta t^2), \tag{4.9}$$

since

$$\mathscr{E}\{[\delta\beta_t]^n\} = \begin{cases} 0, & \text{all odd } n \geqslant 1 \\ 1\cdot 3\cdot 5\cdots(n-1)(\sigma\,\delta t^{1/2})^n, & \text{all even } n \geqslant 2, \end{cases} \tag{4.10}$$

as we have seen in Example 2.6. Thus the representation of (4.6) by the differential equation

$$dx_t = x_t\,d\beta_t$$

involves errors of the order of dt and is consequently meaningless (truncation errors add linearly). Apparently the $\delta\beta_t^2$ term in (4.7) cannot be omitted.

Compute[2]

$$\begin{aligned}&\mathscr{E}\left\{\delta x_t - x_t\,\delta\beta_t - \frac{x_t}{2}\,\delta\beta_t^2\right\} = o(\delta t), \\ &\mathscr{E}\left\{\left|\,\delta x_t - x_t\,\delta\beta_t - \frac{x_t}{2}\,\delta\beta_t^2\,\right|^2\right\} = o(\delta t^2). \end{aligned} \tag{4.11}$$

Therefore, in the mean square sense, we have the differential equation representation

$$dx_t = x_t\,d\beta_t + \tfrac{1}{2}x_t\,d\beta_t^2, \qquad x_0 = 1 \text{ wp } 1. \tag{4.12}$$

Our manipulations with differentials have been purely formal. We have not defined these differentials, and, as a result, we have not given a meaning to (4.12). Integrating (4.12) formally,

$$x_t - 1 = \int_0^t x_\tau\,d\beta_\tau + \frac{1}{2}\int_0^t x_\tau\,d\beta_\tau{}^2. \tag{4.13}$$

[1] $\mathscr{E}\{|\,x_t\,|^n\}$ exists for all n. See Example 3.11. $O(\delta t)$ means that $O(\delta t)/\delta t \to M$ $(\delta t \to 0)$, $0 < M < \infty$.

[2] $o(\delta t)$ means that $o(\delta t)/\delta t \to 0$ $(\delta t \to 0)$.

We shall define the integrals in (4.13). Then the differential dx_t in (4.12) will be considered as a formal notation for (4.13). That is, the differential will be defined in terms of the integral.

Obviously, (4.13) [and consequently (4.12)] should have (4.6) as its solution. Having defined the integrals in (4.13), we shall return to this problem to see that this is so. See Examples 4.3, 4.7, and 4.10.

Example 4.1. We compute the solution of (4.12) by formal manipulations with differentials. Multiply (4.12) by the integrating factor $e^{-\beta t}$:

$$e^{-\beta t}\,dx_t = x_t e^{-\beta t}\,d\beta_t + \tfrac{1}{2}x_t e^{-\beta t}\,d\beta_t{}^2.$$

Now by an expansion procedure similar to (4.7) it is easy to compute the differential

$$d(x_t e^{-\beta t}) = e^{-\beta t}\,dx_t - x_t e^{-\beta t}\,d\beta_t - \tfrac{1}{2}x_t e^{-\beta t}\,d\beta_t{}^2$$

in the mean square sense, where we have used (4.12). Combining these two equations, we get

$$d(x_t e^{-\beta t}) = 0,$$

so that

$$x_t = ce^{\beta t}.$$

In view of the initial condition, $c = 1$, and we get (4.6) as the solution. ▲

3. ITÔ STOCHASTIC INTEGRAL

The integral

$$\int_a^b g(\tau)\,d\beta_\tau\,,$$

where $g(t)$ is a fixed (deterministic) real function and $\{\beta_t\}$ is a scalar Brownian motion process, was introduced by Wiener [18] and is called a *Wiener integral.* This was subsequently generalized by Itô [9] to the case where the function $g_t(\omega)$ is random, and the integral

$$\int_a^b g_\tau(\omega)\,d\beta_\tau$$

will be called the *Itô (stochastic) integral.* In this section, we define the Itô integral and determine some of its properties. Its existence is not rigorously proved, but a plausibility argument is given. The general

mathematical treatment of the Itô integral may be found in Itô [9–11], Doob [6, Chapter IX], Skorokhod [15], or Dynkin [7].

Let $T = [a, b]$, and let $\{\beta_t, t \in T\}$ be a scalar Brownian motion process with variance parameter σ^2. Suppose that the random function $g_t(\omega)$ is defined on T. Partition T:

$$a = t_0 < t_1 < \cdots < t_n = b.$$

First consider *step* functions

$$g_t(\omega) = \begin{cases} 0, & t < t_0 \\ g_i(\omega), & t_i \leqslant t < t_{i+1} \\ 0, & t \geqslant t_n, \end{cases} \tag{4.14}$$

where $g_i(\omega)$ is independent of $\{\beta_{t_k} - \beta_{t_l} : t_i \leqslant t_l \leqslant t_k \leqslant b\}$, and $\mathscr{E}\{|g_i(\omega)|^2\} < \infty$. For such step functions, the Itô integral is defined by

$$\int_T g_t(\omega)\, d\beta_t \triangleq \sum_{i=0}^{n-1} g_i(\omega)(\beta_{t_{i+1}} - \beta_{t_i}). \tag{4.15}$$

We easily see that

$$\mathscr{E}\left\{\int_T g_t(\omega)\, d\beta_t\right\} = 0, \tag{4.16}$$

in view of the independence assumption. Let $f_t(\omega)$ be another step function similarly defined. Then $(j > i)$

$$\mathscr{E}\{g_j(\omega) f_i(\omega)(\beta_{t_{j+1}} - \beta_{t_j})(\beta_{t_{i+1}} - \beta_{t_i})\} = 0,$$

since $(\beta_{t_{j+1}} - \beta_{t_j})$ is independent of the other random variables, and

$$\mathscr{E}\{\beta_{t_{j+1}} - \beta_{t_j}\} = 0.$$

Therefore,

$$\mathscr{E}\left\{\left[\sum_{i=0}^{n-1} g_i(\beta_{t_{i+1}} - \beta_{t_i})\right]\left[\sum_{i=0}^{n-1} f_i(\beta_{t_{i+1}} - \beta_{t_i})\right]\right\}$$

$$= \sigma^2 \sum_{i=0}^{n-1} \mathscr{E}\{g_i f_i\}(t_{i+1} - t_i) = \sigma^2 \int_T \mathscr{E}\{g_t f_t\}\, dt.$$

Thus we have

$$\mathscr{E}\left\{\int_T g_t(\omega)\, d\beta_t \cdot \int_T f_t(\omega)\, d\beta_t\right\} = \sigma^2 \int_T \mathscr{E}\{g_t f_t\}\, dt. \tag{4.17}$$

Now suppose $\{g_t^n\}$ is a sequence of step functions converging to the random function $g_t(\omega)$ in the sense that

$$\int_T \mathscr{E}\{|\, g_t - g_t^n \,|^2\}\, dt \to 0 \qquad (n \to \infty). \tag{4.18}$$

Then, in view of (4.17),

$$\mathscr{E}\left\{\left|\int_T g_t\, d\beta_t - \int_T g_t^n\, d\beta_t\right|^2\right\} = \mathscr{E}\left\{\left|\int_T (g_t - g_t^n)\, d\beta_t\right|^2\right\}$$

$$= \sigma^2 \int_T \mathscr{E}\{|\, g_t - g_t^n \,|^2\}\, dt \to 0 \qquad (n \to \infty).$$

That is,

$$\int_T g_t(\omega)\, d\beta_t = \underset{n\to\infty}{\text{l.i.m.}} \int_T g_t^n\, d\beta_t\,. \tag{4.19}$$

Thus, for all functions $g_t(\omega)$ that can be approximated by step functions in the sense of (4.18), the mean square limit in (4.19) exists. It can be shown (Doob [6, p. 439]) that the class of $g_t(\omega)$ functions satisfying

$$g_t(\omega) \text{ independent of } \{\beta_{t_k} - \beta_{t_l} : t \leqslant t_l \leqslant t_k \leqslant b\}, \qquad \text{for all} \quad t \in T, \tag{4.20}$$

and

$$\int_T \mathscr{E}\{|\, g_t(\omega)|^2\}\, dt < \infty \tag{4.21}$$

can be so approximated. Therefore, for all $g_t(\omega)$ functions satisfying (4.20) and (4.21), the mean square limit in (4.19) exists and defines the **Itô stochastic integral**. In view of Theorem 3.1 (iv), the Itô integral is linear and additive in the domain of integration.

We now show that (4.16) and (4.17) hold in general, since they hold for step function integrands.

Theorem 4.1. *Let the random functions $g_t(\omega)$ and $f_t(\omega)$ satisfy* (4.20) *and* (4.21). *Then their Itô integrals are well defined* [*by* (4.19)], *and*

$$\mathscr{E}\left\{\int_T g_t(\omega)\, d\beta_t\right\} = 0, \tag{4.22}$$

$$\mathscr{E}\left\{\int_T g_t(\omega)\, d\beta_t \cdot \int_T f_t(\omega)\, d\beta_t\right\} = \sigma^2 \int_T \mathscr{E}\{g_t f_t\}\, dt. \tag{4.23}$$

PROOF: (4.22) holds for step function integrands; that is,

$$\mathscr{E}\left\{\int_T g_t^n \, d\beta_t\right\} = 0, \qquad \text{all } n,$$

in view of (4.16). Therefore,

$$\lim_{n\to\infty} \mathscr{E}\left\{\int_T g_t^n \, d\beta_t\right\} = 0.$$

But then, by Theorem 3.1 (v),

$$\mathscr{E}\left\{\text{l.i.m.}_{n\to\infty} \int_T g_t^n \, d\beta_t\right\} = 0,$$

which is (4.22). To prove (4.23), we shall prove that

$$\lim_{n\to\infty} \int_T \mathscr{E}\{g_t^n f_t^n\} \, dt = \int_T \mathscr{E}\{g_t f_t\} \, dt, \tag{4.24}$$

where the sequences of step functions converge according to (4.18). (4.23) will then follow by Theorem 3.1 (vii). Compute

$$\begin{aligned}
&\left|\int_T \mathscr{E}\{g_t^n f_t^n - g_t f_t\} \, dt\right| \\
&\quad = \left|\int_T \mathscr{E}\{g_t^n (f_t^n - f_t) + (g_t^n - g_t) f_t\} \, dt\right| \\
&\quad \leqslant \int_T |\, \mathscr{E}\{g_t^n (f_t^n - f_t)\}| \, dt + \int_T |\, \mathscr{E}\{(g_t^n - g_t) f_t\}| \, dt \\
&\quad \leqslant \int_T \mathscr{E}^{1/2}\{|\, g_t^n \,|^2\} \, \mathscr{E}^{1/2}\{|\, f_t^n - f_t \,|^2\} \, dt + \int_T \mathscr{E}^{1/2}\{|\, g_t^n - g_t \,|^2\} \, \mathscr{E}^{1/2}\{|\, f_t \,|^2\} \, dt \\
&\quad \leqslant \left(\int_T \mathscr{E}\{|\, g_t^n \,|^2\} \, dt\right)^{1/2} \left(\int_T \mathscr{E}\{|\, f_t^n - f_t \,|^2\} \, dt\right)^{1/2} \\
&\qquad + \left(\int_T \mathscr{E}\{|\, g_t^n - g_t \,|^2\} \, dt\right)^{1/2} \left(\int_T \mathscr{E}\{|\, f_t \,|^2\} \, dt\right)^{1/2} \to 0 \qquad (n \to \infty)
\end{aligned}$$

in view of (4.18). ■

Although the definition of the Itô integral is in terms of step functions, we show that under certain conditions it is the mean square limit of Riemann–Stieltjes sums.

Theorem 4.2. *Let* $\mathscr{E}\{|\, g_t(\omega)|^2\} < \infty$ *for all* $t \in T$. *Suppose* $g_t(\omega)$ *is*

mean square continuous on T and satisfies the independence condition (4.20). *Partition* T: $a = t_0 < t_1 < \cdots < t_n = b$, *and let* $\rho = \max_i(t_{i+1} - t_i)$. *Then the Itô stochastic integral equals the following mean square limit*:

$$\text{l.i.m.}_{\rho \to 0} \sum_{i=0}^{n-1} g_{t_i}(\omega)(\beta_{t_{i+1}} - \beta_{t_i}) = \int_T g_t(\omega)\, d\beta_t \,. \tag{4.25}$$

PROOF: By Theorem 3.2, $\mathscr{E}\{|\, g_t(\omega)|^2\}$ is continuous on T. Therefore, condition (4.21) is satisfied, and the Itô integral is well defined. Let

$$\Delta = \int_T g_t(\omega)\, d\beta_t - \sum_{i=0}^{n-1} g_{t_i}(\omega)(\beta_{t_{i+1}} - \beta_{t_i}).$$

Let

$$\tau(t) = t_i \,; \qquad t_i \leqslant t < t_{i+1} \,.$$

Then

$$\Delta = \int_T [g_t(\omega) - g_{\tau(t)}(\omega)]\, d\beta_t \,.$$

By Theorem 4.1,

$$\mathscr{E}\{|\, \Delta\, |^2\} = \sigma^2 \int_T \mathscr{E}\{|\, g_t(\omega) - g_{\tau(t)}(\omega)|^2\}\, dt.$$

Now $\mathscr{E}\{|\, g_t\, |^2\}$ is continuous on T and therefore uniformly bounded on T. Therefore, $\mathscr{E}\{|\, g_t - g_\tau\, |^2\}$ is uniformly bounded on T. Furthermore, as $\rho \to 0$, $\tau \to t$, and therefore

$$\mathscr{E}\{|\, g_t - g_\tau\, |^2\} \to 0 \qquad (\rho \to 0).$$

Therefore, by bounded convergence,

$$\mathscr{E}\{|\, \Delta\, |^2\} \to 0 \qquad (\rho \to 0). \quad \blacksquare$$

In studying stochastic differential equations, we need to consider the Itô integral as a function of its upper limit. Let

$$x_t = \int_a^t g_\tau(\omega)\, d\beta_\tau \,, \qquad t \in T. \tag{4.26}$$

Theorem 4.3. *Conditions of Theorem* 4.2. *Then* x_t *is mean square continuous on* T.

PROOF: In view of Theorems 4.1 and 4.2, $\mathscr{E}\{|\, x_t\,|^2\} < \infty$, $t \in T$. Now, by Theorem 4.1,

$$\mathscr{E}\left\{\left|\int_t^{t+h} g_\tau(\omega)\, d\beta_\tau\right|^2\right\} = \sigma^2 \int_t^{t+h} \mathscr{E}\{|\, g_\tau\,|^2\}\, d\tau \to 0 \qquad (h \to 0),$$

since the (last) Riemann integral is a continuous function of its upper limit. ■

Remark. It can also be shown (Doob [6, p. 445]) that the sample functions of the $\{x_t\}$ process are almost all continuous on T. That is, x_t is continuous wp 1 on T.

In Section 2 we encountered the integral

$$\int_T g_t(\omega)\, d\beta_t{}^2.$$

We now characterize this integral. For our usual partition of the interval T, we define the **second-order stochastic integral**:

$$\int_T g_t(\omega)\, d\beta_t{}^2 \triangleq \operatorname*{l.i.m.}_{\rho \to 0} \sum_{i=0}^{n-1} g_{t_i}(\beta_{t_{i+1}} - \beta_{t_i})^2. \tag{4.27}$$

Theorem 4.4. *Conditions of Theorem* 4.2. *Then*

$$\int_T g_t(\omega)\, d\beta_t{}^2 = \sigma^2 \int_T g_t(\omega)\, dt. \tag{4.28}$$

PROOF: Note that the second integral in (4.28) is a well-defined mean square Riemann integral (Theorem 3.7, Corollary 2). In view of the definitions of the integrals in (4.28), we have to prove that

$$\mathscr{E}\{|\, \Delta\,|^2\} = \mathscr{E}\left\{\left|\sum_{i=0}^{n-1} g_{t_i}[(\beta_{t_{i+1}} - \beta_{t_i})^2 - \sigma^2(t_{i+1} - t_i)]\right|^2\right\}$$

goes to zero with ρ. Now

$$\mathscr{E}\{|\, \Delta\,|^2\} = \mathscr{E}\left\{\sum_{i=0}^{n-1}\sum_{j=0}^{n-1} g_{t_i} g_{t_j}[(\beta_{t_{i+1}} - \beta_{t_i})^2 - \sigma^2(t_{i+1} - t_i)]\right.$$
$$\left. \times\, [(\beta_{t_{j+1}} - \beta_{t_j})^2 - \sigma^2(t_{j+1} - t_j)]\right\}.$$

Consider an $i \neq j$ term in the double sum. Suppose $j > i$. Then

$$[(\beta_{t_{j+1}} - \beta_{t_j})^2 - \sigma^2(t_{j+1} - t_j)] \tag{4.29}$$

is independent of the other rv's in that term. Since the expectation of (4.29) is zero, that term is zero. Therefore,

$$\mathscr{E}\{|\Delta|^2\} = \mathscr{E}\left\{\sum_{i=0}^{n-1} |g_{t_i}|^2 |(\beta_{t_{i+1}} - \beta_{t_i})^2 - \sigma^2(t_{i+1} - t_i)|^2\right\}$$

$$= \sum_{i=0}^{n-1} \mathscr{E}\{|g_{t_i}|^2\}\, \mathscr{E}\{|(\beta_{t_{i+1}} - \beta_{t_i})^2 - \sigma^2(t_{i+1} - t_i)|^2\},$$

in view of the independence condition (4.20). But

$$\mathscr{E}\{|(\beta_{t_{i+1}} - \beta_{t_i})^2 - \sigma^2(t_{i+1} - t_i)|^2\} = 2\sigma^4(t_{i+1} - t_i)^2, \tag{4.30}$$

in view of (4.10). Therefore,

$$\mathscr{E}\{|\Delta|^2\} = 2\sigma^4 \sum_{i=0}^{n-1} \mathscr{E}\{|g_{t_i}|^2\}(t_{i+1} - t_i)^2$$

$$\leqslant 2\sigma^4\rho \sum_{i=0}^{n-1} \mathscr{E}\{|g_{t_i}|^2\}(t_{i+1} - t_i) \to 0 \qquad (\rho \to 0),$$

since

$$\int_T \mathscr{E}\{|g_t|^2\}\, dt < \infty. \quad \blacksquare$$

Example 4.2. One might similarly define a third-order stochastic integral. Let the reader show that

$$\int_T g_t(\omega)\, d\beta_t{}^3 = 0. \quad \blacktriangle$$

Example 4.3. Let us for a moment return to the stochastic differential equation (4.12), which we derived for the $e^{\beta t}$ process in Section 2. In view of Theorem 4.4, (4.13) becomes

$$x_t - 1 = \int_0^t x_\tau\, d\beta_\tau + \sigma^2/2 \int_0^t x_\tau\, d\tau. \tag{4.31}$$

Since, as we have said, the integral defines the differential, (4.12) becomes

$$dx_t = x_t\, d\beta_t + \sigma^2/2\, x_t\, dt, \qquad x_0 = 1 \quad \text{wp 1.} \tag{4.32}$$

Note that $d\beta_t^2$ *is replaced by its expectation,* which is $\sigma^2\,dt$! Note also that the Itô integral in (4.31) apparently cannot be evaluated by the formal rules of integration. For if it could be so evaluated, then $x_t = e^{\beta_t}$ would satisfy

$$x_t - 1 = \int_0^t x_\tau\, d\beta_\tau\,,$$

but, in fact, e^{β_t} satisfies (4.31). ▲

Example 4.4.[3] We evaluate a simple Itô integral. Let $g_t(\omega)$ be the explicit function of β_t,

$$g_t(\omega) = \beta_t - \beta_a\,, \tag{4.33}$$

and evaluate

$$\int_a^b (\beta_t - \beta_a)\, d\beta_t\,.$$

According to Theorem 4.2,

$$\int_a^b (\beta_t - \beta_a)\, d\beta_t = \operatorname*{l.i.m.}_{\rho\to 0} \sum_{i=0}^{n-1} (\beta_{t_i} - \beta_a)(\beta_{t_{i+1}} - \beta_{t_i}),$$

since $g_t(\omega)$ (4.33) satisfies the conditions of that theorem (Section 3.5). Now

$$\sum_{i=0}^{n-1} (\beta_{t_i} - \beta_a)(\beta_{t_{i+1}} - \beta_{t_i}) = \frac{1}{2}(\beta_b - \beta_a)^2 - \frac{1}{2}\sum_{i=0}^{n-1} (\beta_{t_{i+1}} - \beta_{t_i})^2$$

and

$$\begin{aligned}
&\mathscr{E}\left\{\left|\sum_{i=0}^{n-1} (\beta_{t_{i+1}} - \beta_{t_i})^2 - \sigma^2(b-a)\right|^2\right\} \\
&\quad = \mathscr{E}\left\{\left|\sum_{i=0}^{n-1} [(\beta_{t_{i+1}} - \beta_{t_i})^2 - \sigma^2(t_{i+1} - t_i)]\right|^2\right\} \\
&\quad = \mathscr{E}\left\{\sum_{i=0}^{n-1} [(\beta_{t_{i+1}} - \beta_{t_i})^2 - \sigma^2(t_{i+1} - t_i)]^2\right\} \\
&\quad = 2\sigma^4 \sum_{i=0}^{n-1} (t_{i+1} - t_i)^2 \\
&\quad \leqslant 2\sigma^4\rho(b-a) \to 0 \qquad (\rho \to 0).
\end{aligned}$$

[3] This example is due to Doob [6, p. 443].

We used the independence condition (4.20), and (4.30). Therefore,

$$\int_a^b (\beta_t - \beta_a)\, d\beta_t = (1/2)(\beta_b - \beta_a)^2 - (\sigma^2/2)(b - a) \tag{4.34}$$

(see Theorem 3.1). Note that formal rules of integration would give only the first term on the right-hand side of (4.34). ▲

Finally, we note that, if $g_t(\omega)$ is a random n-vector function, then

$$\int_a^b g_t(\omega)\, d\beta_t = \begin{bmatrix} \int_a^b [g_t(\omega)]_1\, d\beta_t \\ \vdots \\ \int_a^b [g_t(\omega)]_n\, d\beta_t \end{bmatrix}.$$

The integrals of the elements of $g_t(\omega)$ have been defined.

4. STOCHASTIC DIFFERENTIAL EQUATIONS

Having defined the Itô stochastic integral, we are in a position to study the stochastic differential equation (4.4):

$$dx_t = f(x_t, t)\, dt + g(x_t, t)\, d\beta_t, \qquad t \in [t_0, T]. \tag{4.35}$$

We suppose for simplicity that (4.35) is a scalar equation and $\{\beta_t, t \geqslant t_0\}$ is a scalar Brownian motion process with unit variance parameter. The assumption of a unit variance parameter is no loss of generality, since $\sigma^2 \neq 1$ can be absorbed in the function g (replace g by σg). By (4.35), we mean the integral equation

$$x_t - x_{t_0} = \int_{t_0}^t f(x_\tau, \tau)\, d\tau + \int_{t_0}^t g(x_\tau, \tau)\, d\beta_\tau, \qquad t \in [t_0, T]. \tag{4.36}$$

The first integral will be a mean square Riemann integral, whereas the second one will be the Itô stochastic integral. Equation (4.35) is therefore called an *Itô stochastic differential equation.*

Theorem 4.5. *Suppose the real functions f and g, and initial condition x_{t_0}, satisfy the hypotheses*:

H_1. *There is a $K > 0$ such that*

$$|f(x, t)| \leqslant K(1 + |x|^2)^{1/2},$$
$$|g(x, t)| \leqslant K(1 + |x|^2)^{1/2}.$$

H_2 .[4] *f and g satisfy uniform Lipschitz conditions in x:*

$$|f(x_2, t) - f(x_1, t)| \leqslant K |x_2 - x_1|,$$
$$|g(x_2, t) - g(x_1, t)| \leqslant K |x_2 - x_1|.$$

H_3 .[5] *f and g satisfy Lipschitz conditions in t on $[t_0, T]$:*

$$|f(x, t_2) - f(x, t_1)| \leqslant K |t_2 - t_1|,$$
$$|g(x, t_2) - g(x, t_1)| \leqslant K |t_2 - t_1|.$$

H_4 . *x_{t_0} is any random variable with $\mathscr{E}\{|x_{t_0}|^2\} < \infty$, independent of $\{d\beta_t, t \in [t_0, T]\}$.*

Then (4.36) *has a solution on $[t_0, T]$ in the mean square sense. The solution $\{x_t\}$ process has the following properties:*

P_1 . *x_t is mean square continuous in $[t_0, T]$.*

P_2 . *$\mathscr{E}\{|x_t|^2\} < \infty$, all $t \in [t_0, T]$.*

P_2' . *$\mathscr{E}\{|x_t|^2\} < M$ on $[t_0, T]$ ($P_2' \Rightarrow P_2$).*

P_3 . *$\int_{t_0}^{T} \mathscr{E}\{|x_t|^2\}\, dt < \infty$.*

P_4 . *$x_t - x_{t_0}$ is independent of $\{d\beta_\tau, \tau \geqslant t\}$ for every $t \in [t_0, T]$.*

Furthermore, the $\{x_t\}$ process is a Markov process, and, in the mean square sense, is uniquely determined by the initial condition x_{t_0} .

The existence proof will be by successive approximations. The following lemma defines the successive approximations.

Lemma 4.1. *Assume hypotheses H_1, H_2, H_3. If an $\{x_t\}$ process has properties P_1, P_2, P_3, and P_4, then the $\{y_t\}$ process defined by*

$$y_t = \int_{t_0}^{t} f(x_\tau, \tau)\, d\tau + \int_{t_0}^{t} g(x_\tau, \tau)\, d\beta_\tau, \qquad t \in [t_0, T], \tag{4.37}$$

has properties P_1, P_2', P_3, and P_4.

PROOF OF LEMMA: By H_1 and P_2,

$$\mathscr{E}\{|f(x_t, t)|^2\} \leqslant K^2(1 + \mathscr{E}\{|x_t|^2\}) < \infty, \qquad \text{all} \quad t \in [t_0, T],$$

[4] Actually, $H_2 \Rightarrow H_1$. See Wong and Zakai [21, p. 7].

[5] H_3 can be weakened to continuity of f and g in t on $[t_0, T]$. See Lemma 3 in Wong and Zakai [21]. However, this complicates the proof.

and similarly for g. By H_2, H_3, and P_1,

$$\begin{aligned}
&\mathcal{E}\{|f(x_{t+h}, t+h) - f(x_t, t)|^2\} \\
&\quad = \mathcal{E}\{|f(x_{t+h}, t+h) - f(x_t, t+h) + f(x_t, t+h) - f(x_t, t)|^2\} \\
&\quad \leqslant 2\mathcal{E}\{|f(x_{t+h}, t+h) - f(x_t, t+h)|^2\} + 2\mathcal{E}\{|f(x_t, t+h) - f(x_t, t)|^2\} \\
&\quad \leqslant 2K^2\mathcal{E}\{|x_{t+h} - x_t|^2\} + 2K^2h^2 \to 0 \qquad (h \to 0),
\end{aligned}$$

and similarly for g. Therefore f and g are ms continuous. By Corollary 2 of Theorem 3.7, f is ms Riemann integrable, and, by Theorem 3.9, the first integral in (4.37) is an ms continuous function of its upper limit. Also, the hypotheses of Theorem 4.2 are satisfied in view of what has been said and P_4, and therefore, by Theorem 4.3, the Itô integral in (4.37) is an ms continuous function of its upper limit. Therefore, $\{y_t\}$ has property P_1. It clearly has property P_4. Furthermore, $P_1 \Rightarrow P_3$ (for y_t) by Theorem 3.2. It remains to demonstrate property P_2'.

$$\begin{aligned}
\mathcal{E}\{|y_t|^2\} &\leqslant 2\mathcal{E}\left\{\left|\int_{t_0}^t f(x_\tau, \tau)\, d\tau\right|^2\right\} + 2\mathcal{E}\left\{\left|\int_{t_0}^t g(x_\tau, \tau)\, d\beta_\tau\right|^2\right\} \\
&= 2\int_{t_0}^t \int_{t_0}^t \mathcal{E}\{f_\tau f_s\}\, d\tau\, ds + 2\int_{t_0}^t \mathcal{E}\{|g_\tau|^2\}\, d\tau \\
&\leqslant 2\left|\int_{t_0}^T \mathcal{E}^{1/2}\{|f_\tau|^2\}\, d\tau\right|^2 + 2\int_{t_0}^T \mathcal{E}\{|g_\tau|^2\}\, d\tau \\
&\leqslant 2(T - t_0)\int_{t_0}^T \mathcal{E}\{|f_\tau|^2\}\, d\tau + 2\int_{t_0}^T \mathcal{E}\{|g_\tau|^2\}\, d\tau \\
&\leqslant 2(T - t_0)K^2\int_{t_0}^T (1 + \mathcal{E}\{|x_\tau|^2\})\, d\tau \\
&\quad + 2K^2\int_{t_0}^T (1 + \mathcal{E}\{|x_\tau|^2\})\, d\tau < M < \infty,
\end{aligned}$$

where the last inequality follows from P_3. ■

PROOF OF THEOREM 4.5: Let $\{x_t^0\}$ be any process with properties P_1, P_2, P_3, and P_4 (for example, $x_t^0 \equiv 0$). We define the successive approximations

$$x_t^n = x_{t_0} + \int_{t_0}^t f(x_\tau^{n-1}, \tau)\, d\tau + \int_{t_0}^t g(x_\tau^{n-1}, \tau)\, d\beta_\tau, \qquad n \geqslant 1. \tag{4.38}$$

According to Lemma 4.1, the $\{x_t^n\}$ process has properties P_1, P_2', P_3, and P_4 for all n. We shall first prove that

$$\text{l.i.m.}_{n\to\infty} x_t^n = x_t, \qquad t \in [t_0, T], \tag{4.39}$$

uniformly in t, and that the $\{x_t\}$ process in (4.39) has properties P_1, P_2', P_3, and P_4. We then show that

$$\text{l.i.m.}_{n\to\infty} \int_{t_0}^{t} f(x_\tau^n, \tau)\, d\tau = \int_{t_0}^{t} f(x_\tau, \tau)\, d\tau, \qquad t \in [t_0, T], \tag{4.40}$$

and

$$\text{l.i.m.}_{n\to\infty} \int_{t_0}^{t} g(x_\tau^n, \tau)\, d\beta_\tau = \int_{t_0}^{t} g(x_\tau, \tau)\, d\beta_\tau, \qquad t \in [t_0, T], \tag{4.41}$$

uniformly in t. (4.40) and (4.41) show that the ms limit in (4.39) is a solution of (4.36). This will prove existence.

We first develop an inequality for later use. Let

$$\begin{aligned} \Delta_n x_t &= x_t^n - x_t^{n-1}, \\ \Delta_n f_t &= f(x_t^n, t) - f(x_t^{n-1}, t), \\ \Delta_n g_t &= g(x_t^n, t) - g(x_t^{n-1}, t). \end{aligned}$$

Then

$$\begin{aligned} \mathcal{E}\{|\Delta_n x_t|^2\} &\leqslant 2\mathcal{E}\left\{\left|\int_{t_0}^{t} \Delta_{n-1} f_\tau\, d\tau\right|^2\right\} + 2\mathcal{E}\left\{\left|\int_{t_0}^{t} \Delta_{n-1} g_\tau\, d\beta_\tau\right|^2\right\} \\ &\leqslant 2(T - t_0)\int_{t_0}^{t} \mathcal{E}\{|\Delta_{n-1} f_\tau|^2\}\, d\tau + 2\int_{t_0}^{t} \mathcal{E}\{|\Delta_{n-1} g_\tau|^2\}\, d\tau \\ &\leqslant 2(T - t_0)K^2\int_{t_0}^{t} \mathcal{E}\{|\Delta_{n-1} x_\tau|^2\}\, d\tau + 2K^2\int_{t_0}^{t} \mathcal{E}\{|\Delta_{n-1} x_\tau|^2\}\, d\tau \\ &= 2K^2(T - t_0 + 1)\int_{t_0}^{t} \mathcal{E}\{|\Delta_{n-1} x_\tau|^2\}\, d\tau. \end{aligned} \tag{4.42}$$

Hence,

$$\mathcal{E}\{|\Delta_2 x_t|^2\} \leqslant 2K^2(T - t_0 + 1)\int_{t_0}^{t} \mathcal{E}\{|\Delta_1 x_\tau|^2\}\, d\tau,$$

$$\begin{aligned} \mathcal{E}\{|\Delta_3 x_t|^2\} &\leqslant [2K^2(T - t_0 + 1)]^2 \int_{t_0}^{t} d\tau \int_{t_0}^{\tau} \mathcal{E}\{|\Delta_1 x_s|^2\}\, ds \\ &= [2K^2(T - t_0 + 1)]^2 \int_{t_0}^{t} (t - \tau)\, \mathcal{E}\{|\Delta_1 x_\tau|^2\}\, d\tau, \end{aligned}$$

where the last equality is obtained by integration by parts. Therefore, by induction,

$$\mathscr{E}\{|\Delta_n x_t|^2\} \leqslant [2K^2(T-t_0+1)]^{n-1}\int_{t_0}^{t}\frac{(t-\tau)^{n-2}}{(n-2)!}\mathscr{E}\{|\Delta_1 x_\tau|^2\}\,d\tau$$

$$\leqslant \frac{C^n}{n!}, \qquad \text{some} \quad C>0, \qquad t\in[t_0, T]. \tag{4.43}$$

We now prove (4.39). For each t and $n > m$,

$$\mathscr{E}\{|x_t^n - x_t^m|^2\} = \mathscr{E}\left\{\left|\sum_{i=m+1}^{n}\Delta_i x_t\right|^2\right\} \leqslant 2^{-m}\sum_{i=m+1}^{n}2^i\mathscr{E}\{|\Delta_i x_t|^2\},$$

and by (4.43),

$$\mathscr{E}\{|x_t^n - x_t^m|^2\} \leqslant 2^{-m}\sum_{i=m+1}^{n}\frac{(2C)^i}{i!} \leqslant 2^{-m}\sum_{1}^{\infty}\frac{(2C)^i}{i!} \to 0 \qquad (m\to\infty). \tag{4.44}$$

Thus by the Cauchy criterion (3.44), the mean square limit in (4.39) exists uniformly in t. By Theorem 3.3, the $\{x_t\}$ process has property P_1. By Theorem 3.2 it has property P_3. It clearly has property P_4. We show that it has property P_2'. When $n\to\infty$ in (4.44), we get

$$\mathscr{E}\{|x_t - x_t^m|^2\} \leqslant 2^{-m}\sum_{1}^{\infty}\frac{(2C)^i}{i!}. \tag{4.45}$$

But

$$|x_t - x_t^m| \geqslant |x_t| - |x_t^m|,$$

so that

$$\mathscr{E}\{|x_t|^2\} \leqslant 2\mathscr{E}\{|x_t^m|^2\} + 2^{1-m}\sum_{1}^{\infty}\frac{(2C)^i}{i!},$$

and $\{x_t\}$ has property P_2', since $\{x_t^m\}$ does.

We next prove (4.40) and (4.41):

$$\mathscr{E}\left\{\left|\int_{t_0}^{t}[f(x_\tau^n,\tau)-f(x_\tau,\tau)]\,d\tau\right|^2\right\} \leqslant (T-t_0)\int_{t_0}^{T}\mathscr{E}\{|f(x_\tau^n,\tau)-f(x_\tau,\tau)|^2\}\,d\tau$$

$$\leqslant (T-t_0)K^2\int_{t_0}^{T}\mathscr{E}\{|x_\tau^n - x_\tau|^2\}\,d\tau$$

$$\leqslant (T-t_0)^2K^2 2^{-n}\sum_{1}^{\infty}\frac{(2C)^i}{i!} \to 0 \quad (n\to\infty).$$

This proves (4.40). (4.41) is proved similarly, using Theorem 4.1

instead of Theorem 3.8 and Schwarz's inequality. This completes the proof of existence.

Let Δ_t be the difference between two solutions of (4.36) with the same initial condition. Then, in the same way we obtained (4.42), we get

$$\mathcal{E}\{|\Delta_t|^2\} \leqslant 2K^2(T - t_0 + 1)\int_{t_0}^{t} \mathcal{E}\{|\Delta_\tau|^2\}\, d\tau. \tag{4.46}$$

Applying (4.46) to itself, we have

$$\mathcal{E}\{|\Delta_t|^2\} \leqslant C^n/n!, \qquad \text{all } n.$$

This proves uniqueness.

Suppose $t, s \in T$, $t \geqslant s$. Then

$$x_t = x_s + \int_s^t f(x_\tau, \tau)\, d\tau + \int_s^t g(x_\tau, \tau)\, d\beta_\tau$$

by the additive property of the integrals. Thus x_t depends on x_s and $\{d\beta_\tau, s \leqslant \tau \leqslant t\}$. By property P_4 and hypothesis H_4, these increments are independent of x_τ, $\tau \leqslant s$. Therefore,

$$p(x_t \mid x_\tau, \tau \leqslant s) = p(x_t \mid x_s),$$

and the $\{x_t\}$ process is Markov. ■

Theorem 4.5 can readily be generalized to the vector case

$$dx_t = f(x_t, t)\, dt + G(x_t, t)\, d\beta_t, \tag{4.47}$$

where x_t and f are n-vectors, G is an $n \times m$ matrix, and β_t is a vector of m independent Brownian motions, each with unit variance parameter. Use the norms

$$|x| = \left(\sum_{i=1}^{n} x_i^2\right)^{1/2} = (x^{\mathrm{T}}x)^{1/2}$$

and

$$|G| = \left(\sum_{i=1}^{n}\sum_{j=1}^{m} G_{ij}^2\right)^{1/2} = [\mathrm{tr}\,(GG^{\mathrm{T}})]^{1/2}.$$

See Section 3.3 for mean square convergence of random vector sequences. We leave this as an exercise for the reader.

It can be shown (Doob [6, p. 277]) that all the conclusions of Theorem 4.5 also hold with probability 1. That is (4.39),

$$\lim_{n\to\infty} x_t^n = x_t \quad \text{wp 1} \tag{4.48}$$

uniformly in t, and [(4.40) and (4.41)]

$$\lim_{n\to\infty} \int_{t_0}^{t} f(x_\tau^n, \tau)\, d\tau = \int_{t_0}^{t} f(x_\tau, \tau)\, d\tau, \quad \text{wp 1}, \tag{4.49}$$

$$\lim_{n\to\infty} \int_{t_0}^{t} g(x_\tau^n, \tau)\, d\beta_\tau = \int_{t_0}^{t} g(x_\tau, \tau)\, d\beta_\tau, \quad \text{wp 1}, \tag{4.50}$$

uniformly in t. Also (uniqueness),

$$\Pr\{\Delta_t = 0,\ t \in [t_0, T]\} = 1. \tag{4.51}$$

That is, (4.36) has a unique solution with probability 1. Furthermore, the x_t sample functions are almost all continuous, and properties P_2', P_3, and P_4 hold. We may note that x_t *does not* have a derivative since the Brownian motion process is not differentiable.

Since, in view of hypotheses H_2 and H_3, $f(\cdot, \cdot)$ is a continuous function of both arguments and x_t is continuous wp 1, f is a continuous function of t wp 1. Therefore $f(x_t, t)$ is Riemann integrable on $[t_0, T]$ wp 1. Thus the first integral in (4.36) is also a well-defined Riemann integral for the sample functions.

Consider a special case of (4.47), the *linear* stochastic differential equation

$$dx_t = F(t)x_t\, dt + G(t)\, d\beta_t, \tag{4.52}$$

where F and G are, respectively, $n \times n$ and $n \times m$ nonrandom matrix functions. In integral form,

$$x_t - x_{t_0} = \int_{t_0}^{t} F(\tau)x_\tau\, d\tau + \int_{t_0}^{t} G(\tau)\, d\beta_\tau. \tag{4.53}$$

Let the initial condition x_{t_0} be Gaussian. Suppose we solve (4.53) by successive approximations (4.38). Let x_t^0 be Gaussian (or $x_t^0 \equiv 0$). Since the integrals in (4.53) are mean square limits, it follows by Theorem 3.11 that x_t^1 is Gaussian. By induction, x_t^n is Gaussian for all n. Since

$$x_t = \text{l.i.m. } x_t^n,$$

x_t is also Gaussian (Theorem 3.11 again). It therefore follows that the process generated by the linear stochastic differential equation (4.52) is a *Gauss–Markov* process.

5. ITÔ STOCHASTIC CALCULUS

We have already seen in Section 2 and in Examples 4.3 and 4.4 that formal integration and differentiation rules cannot be applied to the Itô stochastic integral. In the present section we develop some rules of stochastic calculus, including the stochastic differential and its relationship with the stochastic integral.

Lemma 4.2 (Itô). *Let x_t be the unique solution of the vector Itô stochastic differential equation*

$$dx_t = f(x_t, t)\,dt + G(x_t, t)\,d\beta_t, \qquad t \geqslant t_0, \tag{4.54}$$

where x and f are n-vectors, G is $n \times m$, and $\{\beta_t, t \geqslant t_0\}$ is an m-vector Brownian motion process with $\mathscr{E}\{d\beta_t\, d\beta_t^{\mathrm{T}}\} = Q(t)\,dt$. Let $\varphi(x_t, t)$ be a scalar-valued real function, continuously differentiable in t and having continuous second mixed partial derivatives with respect to the elements of x. Then the (stochastic) differential $d\varphi$ of φ is

$$d\varphi = \varphi_t\,dt + \varphi_x^{\mathrm{T}}\,dx_t + \tfrac{1}{2}\operatorname{tr} GQG^{\mathrm{T}}\varphi_{xx}\,dt, \tag{4.55}$$

where

$$\varphi_t = \frac{\partial \varphi}{\partial t}; \qquad \varphi_x^{\mathrm{T}} = \left[\frac{\partial \varphi}{\partial x_1}, \ \ldots, \ \frac{\partial \varphi}{\partial x_n}\right];$$

$$\varphi_{xx} = \begin{bmatrix} \dfrac{\partial^2 \varphi}{\partial x_1^2}, & \dfrac{\partial^2 \varphi}{\partial x_1\,\partial x_2}, & \cdots, & \dfrac{\partial^2 \varphi}{\partial x_1\,\partial x_n} \\ \vdots & \vdots & & \vdots \\ \dfrac{\partial^2 \varphi}{\partial x_n\,\partial x_1}, & \dfrac{\partial^2 \varphi}{\partial x_n\,\partial x_2}, & \cdots, & \dfrac{\partial^2 \varphi}{\partial x_n^2} \end{bmatrix}.$$

Equation (4.55) is to be understood in the sense that the integral of its right side over $[t_0, T]$ (where the stochastic integral is as defined in Section 3) is $\varphi(T) - \varphi(t_0)$. That is, the differential is defined in terms of the integral. Equation (4.55) tells us what stochastic differential equation a function of the solution of another stochastic differential equation satisfies.

Lemma 4.2 is proved for the scalar case in Itô [11] and Skorokhod [15, p. 24], and the extension to the vector case is straightforward. We shall not prove it here, but give a formal derivation of the result (4.55).

As we did in Section 2, we expand φ in a Taylor's series, retaining second-order terms in $d\beta_t$:

$$\begin{aligned} d\varphi &= \varphi_t\, dt + \varphi_x^{\mathrm{T}}\, dx_t + \tfrac{1}{2}\, dx_t^{\mathrm{T}}\, \varphi_{xx}\, dx_t + \cdots \\ &= \varphi_t\, dt + \varphi_x^{\mathrm{T}}(f\, dt + G\, d\beta_t) + \tfrac{1}{2}\, d\beta_t^{\mathrm{T}}\, G^{\mathrm{T}}\varphi_{xx} G\, d\beta_t + \cdots \\ &= \varphi_t\, dt + \varphi_x^{\mathrm{T}}\, dx_t + \tfrac{1}{2}\,\mathrm{tr}\, G\, d\beta_t\, d\beta_t^{\mathrm{T}}\, G^{\mathrm{T}}\varphi_{xx} + \cdots. \end{aligned}$$

As we saw in Example 4.3, $d\beta_t\, d\beta_t^{\mathrm{T}}$ has to be replaced by its expectation, and so we get

$$d\varphi = \varphi_t\, dt + \varphi_x^{\mathrm{T}}\, dx_t + \tfrac{1}{2}\,\mathrm{tr}\, GQG^{\mathrm{T}}\varphi_{xx}\, dt + o(dt),$$

which is the desired result. Note that (4.56) can be written as

$$d\varphi = (\varphi_t + \varphi_x^{\mathrm{T}} f + \tfrac{1}{2}\,\mathrm{tr}\, GQG^{\mathrm{T}}\varphi_{xx})\, dt + \varphi_x^{\mathrm{T}} G\, d\beta_t\,. \tag{4.56}$$

Example 4.5. In Example 4.4, we saw that

$$\int_a^t (\beta_\tau - \beta_a)\, d\beta_\tau = (1/2)(\beta_t - \beta_a)^2 - (\sigma^2/2)(t - a),$$

where $\{\beta_t\,,\ t \geqslant a\}$ has variance parameter σ^2. Let

$$\varphi = \tfrac{1}{2}(x_t - \beta_a)^2 - (\sigma^2/2)(t - a),$$

and let

$$dx_t = d\beta_t\,.$$

Applying Lemma 4.2, we get

$$\begin{aligned} d\varphi &= -(\sigma^2/2)\, dt + (x_t - \beta_a)\, dx_t + (\sigma^2/2)\, dt \\ &= (\beta_t - \beta_a)\, d\beta_t\,. \end{aligned}$$

Thus, we see that the differential (4.55) is consistent with the definition of the Itô integral. ▲

Corollary 1 (Fundamental Theorem of Itô Stochastic Calculus). *Let $\varphi(x)$ be a twice continuously differentiable real scalar function of the real variable x. Let $\psi = \varphi_x$. Then ($b > a$)*

$$\int_a^b \psi(\beta_t)\, d\beta_t = \varphi(\beta_b) - \varphi(\beta_a) - (\sigma^2/2)\int_a^b \varphi_{xx}(\beta_t)\, dt. \tag{4.57}$$

PROOF: Apply Lemma 4.2 to $\varphi(x_t)$ with $dx_t = d\beta_t$, $\mathscr{E}\{d\beta_t^2\} = \sigma^2\,dt$. We get

$$d\varphi = \varphi_x\,d\beta_t + (\sigma^2/2)\varphi_{xx}\,dt,$$

and integration produces (4.57). ■

If, given ψ, we can find φ such that $\varphi_x = \psi$, we may use (4.57) to evaluate the stochastic integral (in terms of a Riemann integral).

Example 4.6. Evaluate

$$\int_a^b \beta_t^n\,d\beta_t\,.$$

We have $\psi = \beta_t^n$, $\varphi = (n+1)^{-1}\beta_t^{n+1}$, and $\varphi_{\beta\beta} = n\beta_t^{n-1}$. From (4.57),

$$\int_a^b \beta_t^n\,d\beta_t = (n+1)^{-1}(\beta_b^{n+1} - \beta_a^{n+1}) - (n\sigma^2/2)\int_a^b \beta_t^{n-1}\,dt. \tag{4.58}$$

Thus,

$$\int_a^b \beta_t\,d\beta_t = \tfrac{1}{2}(\beta_b^2 - \beta_a^2) - (\sigma^2/2)(b-a),$$

$$\int_a^b \beta_t^2\,d\beta_t = \tfrac{1}{3}(\beta_b^3 - \beta_a^3) - (\sigma^2/2)(\beta_b^2 - \beta_a^2) + 2(\sigma^2/2)^2(b-a),$$

and by induction

$$\int_a^b \beta_t^n\,d\beta_t = (n+1)^{-1}(\beta_b^{n+1} - \beta_a^{n+1}) + \sum_{i=2}^{n}(-1)^{n-i+1}(n!/i!)(\sigma^2/2)^{n-i+1}(\beta_b^i - \beta_a^i)$$
$$+ (-1)^n n!(\sigma^2/2)^n(b-a). \quad ▲ \tag{4.59}$$

Example 4.7. In Section 2 and Example 4.3 we saw that

$$dx_t = x_t\,d\beta_t + (\sigma^2/2)x_t\,dt, \qquad t \geqslant 0, \qquad x_0 = 1,$$

or equivalently,

$$x_t - 1 = \int_0^t x_\tau\,d\beta_\tau + (\sigma^2/2)\int_0^t x_\tau\,d\tau, \tag{4.60}$$

should have the solution

$$x_t = e^{\beta_t}.$$

Let $\psi = e^{\beta_t}$. Then $\varphi = \varphi_{\beta\beta} = e^{\beta_t}$. Then with $b = t$, $a = 0$, Eq. (4.57) becomes

$$\int_0^t e^{\beta_\tau}\, d\beta_\tau = e^{\beta_t} - 1 - (\sigma^2/2)\int_0^t e^{\beta_\tau}\, d\tau \tag{4.61}$$

($\beta_0 = 0$ by definition). Comparing (4.60) with (4.61), we see that e^{β_t} is indeed the solution. ▲

Example 4.8. We saw in Example 4.7 that the solution of

$$dx_t = x_t\, d\beta_t + (\sigma^2/2)x_t\, dt, \qquad t \geqslant 0, \qquad x_0 = 1,$$

is

$$x_t = e^{\beta_t}.$$

What is the solution of

$$dx_t = x_t\, d\beta_t\,, \qquad t \geqslant 0, \qquad x_0 = 1\,?$$

It is easy to verify, with the aid of Lemma 4.2, that the solution is

$$x_t = \exp[\beta_t - (\sigma^2/2)t]. \quad ▲$$

Let x_t be the unique solution of the Itô equation (4.54), and let y_t be the unique solution of the Itô equation

$$dy_t = \bar{f}(y_t\,, t)\, dt + \bar{G}(y_t\,, t)\, d\beta_t\,, \qquad t \geqslant t_0\,,$$

where $\{\beta_t\,, t \geqslant t_0\}$, here and in (4.54), is the same Brownian motion process. Then, by considering the joint process

$$dz_t = \begin{bmatrix} f \\ \bar{f} \end{bmatrix} dt + \begin{bmatrix} G \\ \bar{G} \end{bmatrix} d\beta_t\,, \qquad z_t = [x_t\,, y_t]^{\mathrm{T}},$$

it is easy to prove the following corollaries of Lemma 4.2.

Corollary 2. *Let the product $\varphi\psi$ of the scalar functions $\varphi(x_t\,, t)$ and $\psi(y_t\,, t)$ satisfy the hypotheses of Lemma* 4.2. *Then*

$$d(\varphi\psi) = \psi\, d\varphi + \varphi\, d\psi + \operatorname{tr} G\bar{Q}\bar{G}^{\mathrm{T}}\psi_y\varphi_x{}^{\mathrm{T}}\, dt.$$

Corollary 3. *Let $\psi(y_t\,, t) > 0$, and let the quotient $\varphi\psi^{-1}$ of the scalar functions $\varphi(x_t\,, t)$ and $\psi(y_t\,, t)$ satisfy the hypotheses of Lemma* 4.2. *Then*

$$d(\varphi\psi^{-1}) = \psi^{-1}\, d\varphi - \varphi\psi^{-2}\, d\psi - \psi^{-2} \operatorname{tr} G\bar{Q}\bar{G}^{\mathrm{T}}\psi_y\varphi_x{}^{\mathrm{T}}\, dt$$
$$+ \varphi\psi^{-3} \operatorname{tr} \bar{G}\bar{Q}\bar{G}^{\mathrm{T}}\psi_y\psi_y{}^{\mathrm{T}}\, dt.$$

6. STOCHASTIC INTEGRAL OF STRATONOVICH

Recently Stratonovich [17] proposed a new "symmetric" definition of a stochastic integral. Let $T = [a, b]$, partition T:

$$a = t_0 < t_1 < \cdots < t_n = b,$$

and let $\rho = \max_i(t_{i+1} - t_i)$. Let the scalar random function $g_t(\omega)$, defined on T, be an explicit function of β_t, and denote it by $g(\beta_t, t)$, where $\{\beta_t, t \in T\}$ is a scalar Brownian motion process with variance parameter σ^2. Then the *Stratonovich stochastic integral* is defined by

$$\int_T g(\beta_t, t)\, d\beta_t \triangleq \operatorname*{l.i.m.}_{\rho \to 0} \sum_{i=0}^{n-1} g\left(\frac{\beta_{t_i} + \beta_{t_{i+1}}}{2}, t_i\right)(\beta_{t_{i+1}} - \beta_{t_i}). \tag{4.62}$$

Equivalently, $g[(\beta_{t_i} + \beta_{t_{i+1}})/2, (t_i + t_{i+1})/2]$ can be used instead of $g[(\beta_{t_i} + \beta_{t_{i+1}})/2, t_i]$ in Eq. (4.62). Gray and Caughey [8] noted that $[g(\beta_{t_i}, t_i) + g(\beta_{t_{i+1}}, t_{i+1})]/2$ is also equivalent. Compare (4.62) with (4.25) of Theorem 4.2.

Stratonovich showed that, if $g(\beta, t)$ is continuous in t and has a continuous partial derivative $\partial g/\partial\beta = g_\beta(\beta, t)$, and further satisfies

$$\int_T \mathscr{E}\{|\, g(\beta_t, t)|^2\}\, dt < \infty,$$

then the mean square limit in (4.62) exists and is related to the Itô integral by

$$\int_T g(\beta_t, t)\, d\beta_t = \int_T g(\beta_t, t)\, d\beta_t + (\sigma^2/2)\int_T g_\beta(\beta_t, t)\, dt, \quad \text{wp 1.} \tag{4.63}$$

The last integral in (4.63) is a well-defined Riemann integral for the sample functions in view of the conditions imposed on g.

We note that the Stratonovich integral is only defined for *explicit* functions of β_t, whereas [see (4.20) and (4.21)] the Itô integral is, for example, defined for functionals on $\{\beta_\tau, \tau \leqslant t\}$. Thus the Stratonovich integral is defined for a much more restricted class of functions than is the Itô integral.

Let us evaluate

$$\int_a^b (\beta_t - \beta_a)\, d\beta_t\,.$$

In view of (4.62), we have to compute the ms limit of

$$\sum_{i=0}^{n-1} [(\beta_{t_i} + \beta_{t_{i+1}})/2 - \beta_a](\beta_{t_{i+1}} - \beta_{t_i}) = \sum_{i=0}^{n-1} (\beta_{t_i} - \beta_a)(\beta_{t_{i+1}} - \beta_{t_i}) + \frac{1}{2} \sum_{i=0}^{n-1} (\beta_{t_{i+1}} - \beta_{t_i})^2.$$

Therefore, from Example 4.4,

$$\oint_a^b (\beta_t - \beta_a)\, d\beta_t = \tfrac{1}{2}(\beta_b - \beta_a)^2. \tag{4.64}$$

Thus we see that this Stratonovich integral can be evaluated by formal rules.

In Example 4.7 we saw that $x_t = e^{\beta t}$ is the solution of the Itô equation

$$dx_t = x_t\, d\beta_t + (\sigma^2/2)x_t\, dt, \qquad t \geqslant 0, \qquad x_0 = 1,$$

or

$$\int_0^t dx_\tau = \int_0^t x_\tau\, d\beta_\tau + (\sigma^2/2) \int_0^t x_\tau\, d\tau. \tag{4.65}$$

Substituting for the Itô integral in (4.65) the Stratonovich integral (4.63), and noting that $x_t = e^{\beta t}$, we get

$$\int_0^t dx_\tau = \oint_0^t x_\tau\, d\beta_\tau - (\sigma^2/2) \int_0^t x_\tau\, d\tau + (\sigma^2/2) \int_0^t x_\tau\, d\tau,$$

so that

$$\int_0^t dx_\tau = \oint_0^t x_\tau\, d\beta_\tau .$$

Therefore, $x_t = e^{\beta t}$ is the solution of the Stratonovich equation (with the stochastic integral understood in the Stratonovich sense)

$$dx_t = x_t\, d\beta_t , \qquad t \geqslant 0, \qquad x_0 = 1. \tag{4.66}$$

We could have obtained (4.66) by differentiating $x_t = e^{\beta t}$ formally as if β_t were continuously differentiable. See Example 4.9.

Example 4.9.[6] In Section 2 we gave a motivation for the Itô stochastic differential equation by modeling the scalar process

$$x_t = e^{\beta t}, \qquad t \geqslant 0, \qquad \beta_0 = 0 \quad \text{wp 1}, \tag{4.6}$$

[6] This example is due to Clark [4].

via a *forward* difference equation, and thus we obtained (4.12) in the mean square sense. In view of the symmetric definition of the Stratonovich integral (4.62), let us model (4.6) via a *central* difference equation. In fact, let us consider the general difference operator

$$\delta^{\lambda} x_t \triangleq x_{t+\lambda\delta t} - x_{t-(1-\lambda)\delta t}, \qquad \delta t > 0.$$

With $\lambda = 1$, this becomes a forward difference; with $\lambda = \frac{1}{2}$, a central difference; and with $\lambda = 0$, a backward difference. Using the usual power series expansion, we get from (4.6)

$$\begin{aligned} \delta^{\lambda} x_t &= e^{\beta_{t+\lambda\delta t}} - e^{\beta_{t-(1-\lambda)\delta t}} \\ &= x_t[\delta\beta_t + (\lambda - \tfrac{1}{2})\,\delta\beta_t^2 + O(\delta\beta_t^3)], \end{aligned}$$

where

$$\delta\beta_t = \beta_{t+\delta t} - \beta_t .$$

Thus for a *central* difference ($\lambda = \frac{1}{2}$), the $\delta\beta_t^2$ term is absent, and we have the Stratonovich equation [for (4.6)]

$$dx_t = x_t \, d\beta_t$$

in the mean square sense. Note, however, that this happy circumstance occurs only for $\lambda = \frac{1}{2}$. ▲

It can, in fact, be shown that the Stratonovich integral and the differential that it defines satisfy all the formal rules of calculus, including integration by parts, variable substitution, the chain rule of differentiation, and so on. In the Stratonovich sense, β_t can be treated as if it were continuously differentiable. As a final example, we note that, in view of (4.63), Eq. (4.57) of Corollary 1 (of Lemma 4.2) becomes

$$\int_a^b \psi(\beta_t) \, d\beta_t = \varphi(\beta_b) - \varphi(\beta_a).$$

At this point the reader might ask the following question. Two different stochastic integrals have been defined. Which one is "right?" The answer is that neither one is right, or wrong. They are merely different definitions. If one wants to model the process e^{β_t} via a stochastic differential equation, then one can equivalently use the Itô equation

$$dx_t = x_t \, d\beta_t + (\sigma^2/2) x_t \, dt \qquad \text{(stochastic integral in Itô sense)},$$

or the Stratonovich equation

$$dx_t = x_t \, d\beta_t \qquad \text{(stochastic integral in Stratonovich sense).}$$

The key question is not which equation (integral) to use, but what process is being modeled. The modeling and interpretation of the resultant differential equation have to be consistent. We shall have more to say on the question of modeling in Section 8. In this book we use the Itô definition of the stochastic integral. Our reasons for doing so are outlined in Section 10.

For the Itô equation (4.35),

$$\text{(I)} \qquad dx_t = f(x_t, t)\, dt + g(x_t, t)\, d\beta_t, \qquad t \in [t_0, T], \tag{4.67}$$

there is an equivalent Stratonovich equation. We develop it here formally; the proof can be found in Stratonovich [17]. Let $g(\beta_t, t)$ in (4.63) be $g(x(\beta_t), t)$. Then

$$g_\beta(x(\beta_t), t) = g_x(x_t, t)\,(dx_t/d\beta_t) = g_x(x_t, t)\, g(x_t, t),$$

in view of (4.67). Therefore (4.63) becomes

$$\int_{t_0}^{t} g(x_\tau, \tau)\, d\beta_\tau = \int_{t_0}^{t} g(x_\tau, \tau)\, d\beta_\tau + \tfrac{1}{2}\int_{t_0}^{t} g(x_\tau, \tau)\, g_x(x_\tau, \tau)\, d\tau. \tag{4.68}$$

As a result, the equivalent Stratonovich equation is

$$\text{(S)} \quad dx_t = [f(x_t, t) - \tfrac{1}{2} g(x_t, t)\, g_x(x_t, t)]\, dt + g(x_t, t)\, d\beta_t, \qquad t \in [t_0, T]. \tag{4.69}$$

(β_t is assumed to have unit variance parameter.) Conversely, in view of (4.68), if

$$\text{(S)} \qquad dx_t = f(x_t, t)\, dt + g(x_t, t)\, d\beta_t \tag{4.70}$$

is a Stratonovich equation, the equivalent Itô equation is

$$\text{(I)} \qquad dx_t = [f(x_t, t) + \tfrac{1}{2} g(x_t, t)\, g_x(x_t, t)]\, dt + g(x_t, t)\, d\beta_t. \tag{4.71}$$

For the vector equation (4.47), the transformation (I $\leftrightarrow$ S) is as follows. The ith component of the vector f is modified by

$$\frac{1}{2} \sum_{k=1}^{n} \sum_{j=1}^{m} G_{kj} \frac{\partial G_{ij}}{\partial x_k}. \tag{4.72}$$

Finally, we note that if the noise coefficient g is a nonrandom time

function, the Itô and Stratonovich integrals are equivalent. Thus the linear equation (4.52) is the same in the Itô and Stratonovich sense. Furthermore, in this case, the stochastic integral can be manipulated by formal rules.

7. EVALUATION OF STOCHASTIC (ITÔ) INTEGRALS

We have seen in Example 4.6 how the fundamental theorem of Itô stochastic calculus (Lemma 4.2, Corollary 1) can be used to evaluate stochastic integrals. The relationship between Itô and Stratonovich integrals (4.63) is also useful since the Stratonovich integral can be manipulated by formal rules. Here we present an important result due to Wong and Zakai [19, 21] which relates the stochastic integral to its polygonal approximations. We shall state their result without proof.

Let $T = [a, b]$, and let $a = t_0 < t_1 < \cdots < t_n = b$ be the usual partition of T, with $\rho = \max_i(t_{i+1} - t_i)$. Let $\{\beta_t, t \in T\}$ be a scalar Brownian motion process with variance parameter σ^2. Consider the polygonal approximations to β_t:

$$\beta_t^\rho = \beta_{t_i}^\rho + \frac{\beta_{t_{i+1}}^\rho - \beta_{t_i}^\rho}{t_{i+1} - t_i}(t - t_i), \qquad t_i \leqslant t \leqslant t_{i+1}. \tag{4.73}$$

Theorem 4.6 (Wong and Zakai). *Let* $\mathscr{E}\{|g(\beta_t, t)|^2\} \leqslant M$ *on* T, *and suppose* $g(\beta_t, t)$ *is mean square continuous on* T. *Furthermore, suppose* $g(\beta, t)$ *is continuous in* t *and* $g_\beta(\beta, t)$ *is continuous in* β *and* t, *and* $|g_\beta(\beta, t)| \leqslant K$ *on* T. *Then*

$$\operatorname*{l.i.m.}_{\rho\to 0} \int_a^b g(\beta_t^\rho, t)\, d\beta_t^\rho = \int_a^b g(\beta_t, t)\, d\beta_t + (\sigma^2/2)\int_a^b g_\beta(\beta_t, t)\, dt, \tag{4.74}$$

and, as a consequence of (4.63),

$$\operatorname*{l.i.m.}_{\rho\to 0} \int_a^b g(\beta_t^\rho, t)\, d\beta_t^\rho = \int_a^b g(\beta_t, t)\, d\beta_t. \tag{4.75}$$

Now the polygonal approximations (4.73) are piecewise differentiable, and, as a result of this and the conditions imposed on g, the sequence

$$\left\{\int_a^b g(\beta_t^\rho, t)\, d\beta_t^\rho\right\}$$

is a sequence of ordinary integrals for the sample functions. Therefore, Theorem 4.6 enables us to evaluate the stochastic integral [Itô integral in (4.74) or Stratonovich integral in (4.75)] in terms of ordinary integrals.

Such polygonal approximations are natural physical approximations to the Brownian motion process, which is a mathematical abstraction. (The Brownian motion process is differentiable nowhere, whereas physical Brownian motion is.) It is therefore interesting to note that these approximations converge to the Stratonovich integral, which indicates that this integral may better model a physical process than the Itô integral. We deal with this subject in more detail in the next section.

Example 4.10. For each ρ, the integral

$$\int_a^b \beta_t^\rho \, d\beta_t^\rho = \tfrac{1}{2}(\beta_b^2 - \beta_a^2),$$

and, as a result, so does its ms limit. Compare this with Example 4.6. ▲

Example 4.11. For each ρ, the integral

$$\int_a^b \exp(\beta_t^\rho) \, d\beta_t^\rho = e^{\beta_b} - e^{\beta_a},$$

so that

$$\text{l.i.m.}_{\rho \to 0} \int_a^b \exp(\beta_t^\rho) \, d\beta_t^\rho = e^{\beta_b} - e^{\beta_a} = \oint_a^b e^{\beta_t} \, d\beta_t$$

$$= \int_a^b e^{\beta_t} \, d\beta_t + (\sigma^2/2) \int_a^b e^{\beta_t} \, dt, \tag{4.76}$$

in view of (4.74) and (4.75). Therefore, with $\beta_0 = 0$ wp 1,

$$e^{\beta_t} - 1 = \int_0^t e^{\beta_\tau} \, d\beta_\tau + (\sigma^2/2) \int_0^t e^{\beta_\tau} \, d\tau = \oint_0^t e^{\beta_\tau} \, d\beta_\tau, \tag{4.77}$$

and again we see that the solution of the Itô equation (4.32) is e^{β_t}, which is also the solution of the Stratonovich equation

$$\text{(S)} \qquad dx_t = x_t \circ d\beta_t. \quad ▲$$

8. MORE ON MODELING

We have already noted in Sections 4 and 7 that solutions of stochastic differential equations (be they of the Itô or Stratonovich type) have no derivatives. This is, of course, a result of the erratic properties of the

It is clear that if $\{y_t\}$ is a vector physical process, say of dimension n, then we need at least a $2n$-dimensional system of stochastic differential equations to model that physical process by the method we have just described. We are therefore forced to deal with a process of high dimension.

Is it possible, in some sense, to model the physical process $\{y_t\}$ by a stochastic differential equation of the same dimension? This is an area in which much theoretical and experimental research remains to be done. Some interesting results have been obtained by Clark [4] and Wong and Zakai [19–21]. A detailed treatment of this subject is outside the scope of this book, and we present only a simple example due to W. M. Wonham (see Kushner [13]).

Consider a special case of the physical process (4.78), namely,

$$\dot{y}_t = y_t m_t, \qquad t \geqslant 0, \qquad y_0 = 1. \tag{4.86}$$

Equation (4.86) has the solution

$$y_t^\rho = \exp\left(\int_0^t m_\tau \, d\tau\right), \tag{4.87}$$

where the superscript ρ indicates that m_t comes from (4.80) for a fixed ρ. Compute

$$\mathscr{E}\left\{\left|\int_0^t m_\tau \, d\tau - \sigma\beta_t\right|^2\right\} = \int_0^t \int_0^t \mathscr{E}\{m_\tau m_s\} \, d\tau \, ds - 2\sigma\mathscr{E}\left\{\beta_t \int_0^t m_\tau \, d\tau\right\} + \sigma^2 t.$$

With the aid of (4.85), it is easy to compute

$$\int_0^t \int_0^t \mathscr{E}\{m_\tau m_s\} \, d\tau \, ds = \sigma^2 t + (\sigma^2/\rho)(e^{-\rho t} - 1).$$

From (4.80),

$$-2\sigma\mathscr{E}\left\{\beta_t \int_0^t m_\tau \, d\tau\right\} = (2t/\rho)\,\mathscr{E}\{(m_t - m_0)\} - 2\sigma^2 t = -2\sigma^2 t.$$

Therefore,

$$\mathscr{E}\left\{\left|\int_0^t m_\tau \, d\tau - \sigma\beta_t\right|^2\right\} = (\sigma^2/\rho)(e^{-\rho t} - 1) \to 0 \qquad (\rho \to \infty).$$

In other words,

$$\underset{\rho\to\infty}{\text{l.i.m.}} \int_0^t m_\tau \, d\tau = \sigma\beta_t. \tag{4.88}$$

Therefore, by Example 3.11,

$$\text{l.i.m.}_{\rho\to\infty} y_t^{\rho} = e^{\sigma\beta_t}. \tag{4.89}$$

But $e^{\sigma\beta_t}$ is the solution of the Stratonovich equation

$$\text{(S)} \qquad dx_t = x_t \, d\bar{\beta}_t \tag{4.90}$$

and the Itô equation

$$\text{(I)} \qquad dx_t = x_t \, d\bar{\beta}_t + (\sigma^2/2)x_t \, dt, \tag{4.91}$$

where $\{\bar{\beta}_t\}$ has variance parameter σ^2 (see Section 6 and Example 4.11). Therefore, for ρ large, (4.86) can be modeled by (4.90) or (4.91).

We present, without proof, an important related result due to Wong and Zakai [19]. Consider a sequence of differential equations

$$dx_t^{\rho} = f(x_t^{\rho}, t) \, dt + g(x_t^{\rho}, t) \, d\beta_t^{\rho}, \qquad t \in [a, b] = T, \tag{4.92}$$

where β_t^{ρ} is a sequence of polygonal approximations to β_t, given in (4.73). Let $x_a^{\rho} = x_a$. Since the polygonal approximations are piecewise differentiable wp 1, Eqs. (4.92) are ordinary differential equations for the sample functions.

Theorem 4.7 (Wong and Zakai).[8] *Let f and g satisfy the hypotheses of Theorem* 4.5, *and in addition, let $g_x(x, t)$ be continuous in x and t, and let $g(x, t)\, g_x(x, t)$ satisfy a uniform Lipschitz condition in x and be continuous in t. Furthermore, suppose $\mathscr{E}\{| x_a |^4\} < \infty$. Then*

$$\text{l.i.m.}_{\rho\to 0} x_t^{\rho} = x_t, \tag{4.93}$$

where x_t is the unique solution of the Stratonovich equation

$$\text{(S)} \qquad dx_t = f(x_t, t) \, dt + g(x_t, t) \, d\beta_t \tag{4.94}$$

and the Itô equation

$$\text{(I)} \qquad dx_t = [f(x_t, t) + \tfrac{1}{2} g(x_t, t)\, g_x(x_t, t)] \, dt + g(x_t, t) \, d\beta_t. \tag{4.95}$$

This theorem is useful in the numerical simulation of stochastic differential equations. In this connection, see also Example 4.9.

[8] A related result for more general approximations to the Brownian motion process is given by Wong and Zakai [20].

In general, consider the vector physical process

$$\dot{y}_t = f(y_t, t) + G(y_t, t)m_t, \tag{4.96}$$

where the components of m_t are *independent*. Let m_t be a wide-band or rapidly fluctuating but still continuous Gaussian process and let

$$\text{l.i.m.} \int_0^t m_\tau \, d\tau = \beta_t, \tag{4.97}$$

where β_t is a vector process of independent Brownian motions with unit variance parameters. Then (see Clark [4]) in the mean square sense, (4.96) is modeled by the Stratonovich equation

$$\text{(S)} \qquad dx_t = f(x_t, t)\, dt + G(x_t, t)\, d\beta_t \tag{4.98}$$

or the equivalent Itô equation

$$\text{(I)} \qquad (dx_t)_i = \left[f_i(x_t, t) + \frac{1}{2}\sum_{j,k} G_{kj}\frac{\partial G_{ij}}{\partial x_k}\right] dt + \sum_j G_{ij}\,(d\beta_t)_j. \tag{4.99}$$

These results are in general not valid if the components of m_t are not independent (see Clark [4] for explicit conditions and assumptions).

In connection with modeling, the reader might also consult Stratonovich [16], Kulman [12], Astrom [1], and Gray and Caughey [8].

9. KOLMOGOROV'S EQUATIONS

As was proved in Theorem 4.5, the $\{x_t, t \in [t_0, T]\}$ process generated by the Itô stochastic differential equation (4.35),

$$dx_t = f(x_t, t)\, dt + g(x_t, t)\, d\beta_t, \qquad t \in [t_0, T], \tag{4.100}$$

is a Markov process. [As in Section 4, we assume without loss of generality that $\{\beta_t\}$ has unit variance parameter. Unless otherwise specified, (4.100) is understood in the Itô sense.] The $\{x_t\}$ process is therefore characterized by the density function

$$p(x_t) = p(x, t), \qquad \text{all} \quad t \in [t_0, T], \tag{4.101}$$

and the transition probability density function

$$p(x_t \mid x_\tau) = p_{x_t \mid x_\tau}(x \mid y) = p(x, t; y, \tau), \qquad \text{all} \quad t > \tau \in [t_0, T]. \tag{4.102}$$

See Section 3.7 on Markov processes, and Section 3.2 for the notation.

In this section, we formally derive the laws of evolution of the densities in (4.101) and (4.102), assuming the existence of the continuous partial derivatives

$$\frac{\partial p}{\partial t}, \qquad \frac{\partial [pf(x,t)]}{\partial x}, \qquad \frac{\partial^2 [pg^2(x,t)]}{\partial x^2},$$

where p stands for $p(x, t)$ and $p(x, t; y, \tau)$. In the following, $\delta(\cdot)$ stands for a forward difference

$$\delta\xi = \xi_{t+\delta t} - \xi_t, \qquad \delta t > 0.$$

We consider the transition probability density function first.

Let $R(x)$ be a nonnegative, twice continuously differentiable real function such that $(x_1 < x_2)$

$$R(x) = 0, \qquad x < x_1 \quad \text{and} \quad x > x_2,$$
$$R(x_1) = R(x_2) = R'(x_1) = R'(x_2) = R''(x_1) = R''(x_2) = 0,$$

where the prime denotes the derivative. Then

$$\int_{x_1}^{x_2} \left[\frac{\partial p(x, t; y, \tau)}{\partial t}\,\delta t + o(\delta t)\right] R(x)\,dx$$
$$= \int_{-\infty}^{\infty} [p(x, t + \delta t; y, \tau) - p(x, t; y, \tau)]\,R(x)\,dx. \tag{4.103}$$

Now according to the Chapman–Kolmogorov equation (3.98),

$$p(x, t + \delta t; y, \tau) = \int_{-\infty}^{\infty} p(x, t + \delta t; z, t)\,p(z, t; y, \tau)\,dz. \tag{4.104}$$

Using (4.104), the right-hand side of (4.103) becomes

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x, t + \delta t; z, t)\,p(z, t; y, \tau)\,R(x)\,dz\,dx - \int_{-\infty}^{\infty} p(x, t; y, \tau)\,R(x)\,dx$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(z, t + \delta t; x, t)\,p(x, t; y, \tau)\,R(z)\,dx\,dz$$
$$- \int_{-\infty}^{\infty} p(x, t; y, \tau)\,R(x)\,dx$$
$$= \int_{-\infty}^{\infty} p(x, t; y, \tau)\left\{\int_{-\infty}^{\infty} p(z, t + \delta t; x, t)\,R(z)\,dz - R(x)\right\} dx, \tag{4.105}$$

where we replaced the dummy variable of integration x by z, and z by x.

We next reduce the term in braces in (4.105). Expand $R(z)$ in Taylor's series:

$$R(z) = R(x) + (z - x)\,R'(x) + \tfrac{1}{2}(z - x)^2\,R''(x) + o(z - x)^2.$$

Then

$$\begin{aligned}\int_{-\infty}^{\infty} p(z, t + \delta t; x, t)\,R(z)\,dz - R(x)\\ = R(x)\int_{-\infty}^{\infty} p(z, t + \delta t; x, t)\,dz + R'(x)\int_{-\infty}^{\infty} (z - x)\,p(z, t + \delta t; x, t)\,dz\\ + \tfrac{1}{2}R''(x)\int_{-\infty}^{\infty} (z - x)^2\,p(z, t + \delta t; x, t)\,dz\\ + \int_{-\infty}^{\infty} o(z - x)^2\,p(z, t + \delta t; x, t)\,dz - R(x). \qquad (4.106)\end{aligned}$$

Now

$$\int_{-\infty}^{\infty} p(z, t + \delta t; x, t)\,dz = 1, \qquad (4.107)$$

since the integrand is a density function. Notice that, in (4.106), z is a realization of $x_{t+\delta t}$ and x is a realization of x_t. Therefore, $z - x = \delta x$ is a realization of δx_t. As a result,

$$\begin{aligned}p(z, t + \delta t; x, t) = p_{x_{t+\delta t}|x_t}(z \mid x) = p_{x_{t+\delta t}|x_t}(x + \delta x \mid x)\\ = p_{\delta x_t|x_t}(\delta x \mid x). \qquad (4.108)\end{aligned}$$

Consequently,

$$\int_{-\infty}^{\infty} (z - x)^k\,p(z, t + \delta t; x, t)\,dz = \mathscr{E}\{\delta x_t{}^k \mid x_t\}, \qquad k = 1, 2, \ldots. \qquad (4.109)$$

But, in view of (4.100) and (4.10),

$$\begin{aligned}\mathscr{E}\{\delta x_t \mid x_t\} &= f(x, t)\,\delta t,\\ \mathscr{E}\{\delta x_t{}^2 \mid x_t\} &= g^2(x, t)\,\delta t + o(\delta t), \qquad (4.110)\\ \mathscr{E}\{\delta x_t{}^k \mid x_t\} &= o(\delta t), \qquad k > 2.\end{aligned}$$

Combining (4.109) with (4.110) and using (4.107), (4.106) becomes

$$\begin{aligned}\int_{-\infty}^{\infty} p(z, t + \delta t; x, t)R(z)\,dz - R(x)\\ = R'(x)f(x, t)\,\delta t + \tfrac{1}{2}R''(x)\,g^2(x, t)\,\delta t + o(\delta t). \qquad (4.111)\end{aligned}$$

Therefore, in view of (4.105), (4.103) becomes

$$\int_{x_1}^{x_2} \left[\frac{\partial p(x, t; y, \tau)}{\partial t}\,\delta t + o(\delta t)\right] R(x)\,dx$$

$$= \int_{-\infty}^{\infty} p(x, t; y, \tau)[R'(x) f(x, t)\,\delta t + \tfrac{1}{2}R''(x)\,g^2(x, t)\,\delta t + o(\delta t)]\,dx,$$

and, dividing by δt and taking the $\delta t = 0$ limit, we finally get

$$\int_{x_1}^{x_2} \frac{\partial p(x, t; y, \tau)}{\partial t} R(x)\,dx = \int_{x_1}^{x_2} p(x, t; y, \tau) f(x, t)\,R'(x)\,dx + \frac{1}{2}\int_{x_1}^{x_2} p(x, t; y, \tau)\,g^2(x, t)\,R''(x)\,dx. \quad (4.112)$$

Our derivation is almost complete. We evaluate the integrals on the right-hand side of (4.112) by parts and, using the properties of $R(x)$, obtain

$$\int_{x_1}^{x_2} \left\{\frac{\partial p}{\partial t} + \frac{\partial (pf)}{\partial x} - \frac{1}{2}\frac{\partial^2 (pg^2)}{\partial x^2}\right\} R(x)\,dx = 0. \quad (4.113)$$

Since $R(x)$ is arbitrary, the term in braces must vanish:

$$\frac{\partial p(x, t; y, \tau)}{\partial t} = -\frac{\partial [p(x, t; y, \tau) f(x, t)]}{\partial x} + \frac{1}{2}\frac{\partial^2 [p(x, t; y, \tau)\,g^2(x, t)]}{\partial x^2}. \quad (4.114)$$

The partial differential equation (4.114) is known as **Kolmogorov's forward equation** or the **Fokker–Planck** equation. It describes the evolution of the transition probability density of the Markov process generated by the Itô equation (4.100). A Markov process whose transition probability density satisfies Kolmogorov's equation is called a **diffusion** process.

The initial condition for (4.114) is clearly

$$\lim_{t \to \tau} p_{x_t | x_\tau}(x \mid y) = \delta(x - y). \quad (4.115)$$

If $p(x, t; y, \tau)$ is assumed "well behaved" at infinity, the boundary conditions are

$$p_{x_t | x_\tau}(\infty \mid y) = p_{x_t | x_\tau}(-\infty \mid y) = 0. \quad (4.116)$$

One can also similarly derive the so-called **Kolmogorov's backward equation,** which is

$$-\frac{\partial p(x, t; y, \tau)}{\partial \tau} = f(y, \tau)\frac{\partial p(x, t; y, \tau)}{\partial y} + \tfrac{1}{2}g^2(y, \tau)\frac{\partial^2 p(x, t; y, \tau)}{\partial y^2}. \tag{4.117}$$

The backward equation is the formal adjoint of the forward equation, but it will be of little interest in this book.

In the vector case, that is, for the vector Itô stochastic differential equation (4.47),

$$dx_t = f(x_t, t)\,dt + G(x_t, t)\,d\beta_t \tag{4.118}$$

[x, f are n-vectors, G is $n \times m$, $\{\beta_t\}$ is an m-vector Brownian motion process with $\mathscr{E}\{d\beta_t\, d\beta_t^{\mathrm{T}}\} = Q(t)\,dt$], Kolmogorov's forward equation becomes

$$\frac{\partial p}{\partial t} = -\sum_{i=1}^{n}\frac{\partial (pf_i)}{\partial x_i} + \frac{1}{2}\sum_{i,j=1}^{n}\frac{\partial^2[p(GQG^{\mathrm{T}})_{ij}]}{\partial x_i\,\partial x_j}. \tag{4.119}$$

Its derivation is completely analogous to that of the scalar equation and is left for the reader.

Recall that $p(x, t; y, \tau)$ is a random variable since it depends on the values y taken on by x_τ. With this in mind, we take the expectation of (4.114) and, interchanging expectation with differentiation, obtain

$$\frac{\partial p(x, t)}{\partial t} = -\frac{\partial [p(x, t) f(x, t)]}{\partial x} + \frac{1}{2}\frac{\partial^2 [p(x, t)\, g^2(x, t)]}{\partial x^2}. \tag{4.120}$$

Thus we see that $p(x_t)$ also satisfies Kolmogorov's forward equation. The initial condition for (4.120) is clearly $p(x_{t_0})$, which is assumed given.

Kolmogorov's forward equation (4.119) can be written as

$$dp = \mathscr{L}(p)\,dt, \tag{4.121}$$

were $\mathscr{L}$ is called the *forward diffusion operator* of the $\{x_t\}$ process generated by the Itô equation (4.118):

$$\mathscr{L}(\cdot) = -\sum_{i=1}^{n}\frac{\partial(\cdot f_i)}{\partial x_i} + \frac{1}{2}\sum_{i,j=1}^{n}\frac{\partial^2[\cdot\,(GQG^{\mathrm{T}})_{ij}]}{\partial x_i\,\partial x_j}. \tag{4.122}$$

Example 4.13. Kolmogorov's forward equation has been solved only

in a few simple cases. Some of these solutions may be found in Bharucha–Reid [2]. Using Laplace transform techniques, let the reader solve the Kolmogorov's forward equation:

$$\frac{\partial p(x, t)}{\partial t} = \frac{1}{2}\frac{\partial^2 p(x, t)}{\partial x^2}, \qquad t \geqslant 0, \qquad p(x, 0) = \delta(x).$$

This is the Kolmogorov's equation for the Itô equation

$$dx_t = d\beta_t, \qquad t \geqslant 0, \qquad \beta_0 = 0 \quad \text{wp 1},$$

and thus clearly must have the solution

$$p(x, t) = (1/2\pi t)^{1/2} \exp(-x^2/2t). \quad \blacktriangle$$

Example 4.14. Suppose the stochastic differential equation (4.118) is interpreted in the Stratonovich sense. Then (see Sect. 6) the equivalent Itô equation is

$$d(x_t)_i = \left[f_i(x_t, t) + \frac{1}{2}\sum_{k=1}^{n}\sum_{j=1}^{m}\sum_{l=1}^{m} G_{kj}Q_{jl}\frac{\partial G_{il}}{\partial x_k}\right] dt$$

$$+ \sum_{j=1}^{m} G_{ij}\, d(\beta_t)_j. \tag{4.123}$$

Substituting the quantity in brackets for f_i in (4.119), we get

$$\frac{\partial p}{\partial t} = -\sum_{i=1}^{n}\frac{\partial (pf_i)}{\partial x_i} + \frac{1}{2}\sum_{i,k=1}^{n}\sum_{j=1}^{m}\frac{\partial}{\partial x_i}\left[(GQ)_{ij}\frac{\partial}{\partial x_k}(pG_{kj})\right]. \tag{4.124}$$

This is Kolmogorov's forward equation for the *Stratonovich* equation

$$\text{(S)} \qquad dx_t = f(x_t, t)\, dt + G(x_t, t)\, d\beta_t. \tag{4.125}$$

One can similarly obtain Kolmogorov's backward equation for (4.125). ▲

Consider the *linear* Itô stochastic differential equation

$$dx_t = F(t)x_t\, dt + G(t)\, d\beta_t, \qquad t \geqslant t_0, \tag{4.126}$$

where x_t is an n-vector, F and G are, respectively, $n \times n$ and $n \times m$ time matrix functions, and $\{\beta_t, t \geqslant t_0\}$ is an m-vector Brownian motion process with

$$\mathcal{E}\{d\beta_t\, d\beta_t^{\mathrm{T}}\} = Q(t)\, dt. \tag{4.127}$$

Let $x_{t_0} \sim N(\hat{x}_{t_0}, P_{t_0})$, independent of $\{\beta_t, t \geqslant t_0\}$. We saw in Section 4 that the $\{x_t, t \geqslant t_0\}$ process generated by (4.126) is Gauss–Markov. As a result, $p(x_t)$ is completely characterized by its mean and covariance matrix

$$\hat{x}_t = \mathscr{E}\{x_t\}, \tag{4.128}$$

$$P_t = \mathscr{E}\{(x_t - \hat{x}_t)(x_t - \hat{x}_t)^{\mathrm{T}}\}. \tag{4.129}$$

In the next four examples we derive the equations of evolution for $\hat{x}_t$ and P_t by four different methods.

Example 4.15. For the linear system (4.126), Kolmogorov's forward equation (4.119) becomes

$$\partial p/\partial t = -p \operatorname{tr} F - p_x^{\mathrm{T}} F x + \tfrac{1}{2} \operatorname{tr} GQG^{\mathrm{T}} p_{xx}\,. \tag{4.130}$$

Note that in this linear case (4.124) also reduces to (4.130). This is consistent with the fact that (Section 6) the linear system (4.126) is the same in the Itô and Stratonovich sense.

Recall that the characteristic function of x_t is just the Fourier transform of $p(x, t)$ (2.71):

$$\varphi(u, t) = \int_{-\infty}^{\infty} e^{iu^{\mathrm{T}}x} p(x, t)\, dx. \tag{4.131}$$

Taking the Fourier transform in (4.130),

$$\begin{aligned} \partial\varphi/\partial t &= \int e^{iu^{\mathrm{T}}x}(\partial p/\partial t)\, dx = -\varphi \operatorname{tr} F - \int e^{iu^{\mathrm{T}}x} p_x^{\mathrm{T}} F x\, dx \\ &\quad + \tfrac{1}{2} \operatorname{tr} GQG^{\mathrm{T}} \int e^{iu^{\mathrm{T}}x} p_{xx}\, dx \\ &= -\varphi \operatorname{tr} F - \operatorname{tr} F \int x p_x^{\mathrm{T}} e^{iu^{\mathrm{T}}x}\, dx + \tfrac{1}{2} \operatorname{tr} GQG^{\mathrm{T}} \int p_{xx} e^{iu^{\mathrm{T}}x}\, dx. \end{aligned} \tag{4.132}$$

Evaluating the last two integrals by parts and assuming that p is well behaved at infinity (p and its partial derivatives with respect to x vanish at infinity), we get

$$\partial\varphi/\partial t = u^{\mathrm{T}} F \varphi_u - \tfrac{1}{2}\varphi u^{\mathrm{T}} GQG^{\mathrm{T}} u, \tag{4.133}$$

which is a partial differential equation for the characteristic function.

Since x_t is Gaussian, we know that its characteristic function is simply (2.130)

$$\varphi(u, t) = \exp(iu^{\mathrm{T}}\hat{x}_t - \tfrac{1}{2} u^{\mathrm{T}} P_t u). \tag{4.134}$$

Differentiating (4.134),

$$\partial\varphi/\partial t = \varphi[iu^{\mathrm{T}}(d\hat{x}_t/dt) - \tfrac{1}{2}u^{\mathrm{T}}(dP_t/dt)u], \tag{4.135}$$

$$\varphi_u = \varphi[i\hat{x}_t - P_t u], \tag{4.136}$$

and substituting in (4.133),

$$\begin{aligned}\varphi iu^{\mathrm{T}}(d\hat{x}_t/dt) - \tfrac{1}{2}\varphi u^{\mathrm{T}}(dP_t/dt)u \\ = \varphi iu^{\mathrm{T}}F\hat{x}_t - \tfrac{1}{2}\varphi u^{\mathrm{T}}(FP_t + P_tF^{\mathrm{T}})u - \tfrac{1}{2}\varphi u^{\mathrm{T}}GQG^{\mathrm{T}}u.\end{aligned} \tag{4.137}$$

(We symmetrized FP_t, since its skew-symmetric part vanishes in the quadratic form.)

Now equating the real and imaginary parts in (4.137), we get

$$\begin{aligned}d\hat{x}_t/dt &= F(t)\hat{x}_t\,, \\ dP_t/dt &= F(t)P_t + P_tF^{\mathrm{T}}(t) + G(t)Q(t)\,G^{\mathrm{T}}(t),\end{aligned} \tag{4.138}$$

which are ordinary differential equations describing the evolution of $\hat{x}_t$ and P_t. Initial conditions for (4.138) are given, namely, $\hat{x}_{t_0}$ and P_{t_0}. ▲

Example 4.16. We now obtain (4.138) directly from (4.126). Taking the expectation in (4.126), and interchanging d and $\mathscr{E}$, we get

$$d\hat{x}_t = F(t)\hat{x}_t\,dt, \tag{4.139}$$

which is the first of (4.138) ($\mathscr{E}\{d\beta_t\} \equiv 0$). Now

$$dP_t = d[\mathscr{E}\{x_tx_t^{\mathrm{T}}\} - \hat{x}_t\hat{x}_t^{\mathrm{T}}] = \mathscr{E}\{d(x_tx_t^{\mathrm{T}})\} - d(\hat{x}_t\hat{x}_t^{\mathrm{T}}). \tag{4.140}$$

Applying Itô's lemma 4.2 to the elements of $x_tx_t^{\mathrm{T}}$,

$$d(x_tx_t^{\mathrm{T}}) = x_t\,dx_t^{\mathrm{T}} + (dx_t)x_t^{\mathrm{T}} + GQG^{\mathrm{T}}\,dt, \tag{4.141}$$

and a trivial application of Itô's lemma gives

$$d(\hat{x}_t\hat{x}_t^{\mathrm{T}}) = \hat{x}_t\,d\hat{x}_t^{\mathrm{T}} + (d\hat{x}_t)\hat{x}_t^{\mathrm{T}}, \tag{4.142}$$

since the differential equation for $\hat{x}_t$ (4.139) is not stochastic. Now using (4.126) and (4.139),

$$\begin{aligned}dP_t = \mathscr{E}\{x_t(x_t^{\mathrm{T}}F^{\mathrm{T}}\,dt + d\beta_t^{\mathrm{T}}G^{\mathrm{T}}) + (Fx_t\,dt + G\,d\beta_t)x_t^{\mathrm{T}} + GQG^{\mathrm{T}}\,dt\} \\ - \hat{x}_t\hat{x}_t^{\mathrm{T}}F^{\mathrm{T}}\,dt - F\hat{x}_t\hat{x}_t^{\mathrm{T}}\,dt \\ (\mathscr{E}\{x_tx_t^{\mathrm{T}}\} - \hat{x}_t\hat{x}_t^{\mathrm{T}})F^{\mathrm{T}}\,dt + F(\mathscr{E}\{x_tx_t^{\mathrm{T}}\} - \hat{x}_t\hat{x}_t^{\mathrm{T}})\,dt \\ + GQG^{\mathrm{T}}\,dt,\end{aligned} \tag{4.143}$$

since $\mathscr{E}\{x_t\,d\beta_t^{\mathrm{T}}\} = 0$. (4.143) is just the second of (4.138). ▲

We used Itô's stochastic calculus in Example 4.16, namely, Lemma 4.2, to derive (4.138). Since (4.126) is linear, and the stochastic integral in that equation is therefore the same in the Itô and Stratonovich sense, we could use formal rules of differentiation to obtain the same result. The GQG^{T} term in (4.141) would then be absent, but the differential operator d in that equation would have to be defined as a limit of a *central* difference, rather than the usual limit of a forward difference (see Example 4.9). Defined this way, the first two terms on the right-hand side of (4.141) themselves contribute a GQG^{T} term since

$$\mathcal{E}\{x_t\, d^{1/2}\beta_t^{\mathrm{T}}\} \neq 0$$

if $d^{1/2}$ is a "central difference" differential. We leave the details as an exercise for the reader.

Example 4.17. We saw in Section 6 that the linear equation (4.126) can be manipulated by formal rules. Consider, therefore, the equivalent equation

$$\dot{x}_t = F(t)x_t + G(t)w_t\,, \qquad t \geqslant t_0\,, \tag{4.144}$$

where $\{w_t\,,\ t \geqslant t_0\}$ is a zero-mean white Gaussian noise process with

$$\mathcal{E}\{w_t w_\tau^{\mathrm{T}}\} = Q(t)\,\delta(t-\tau).$$

The solution of (4.144) is

$$x_t = \Phi(t, t_0)x_{t_0} + \int_{t_0}^{t} \Phi(t, \tau)\, G(\tau)w_\tau\, d\tau, \tag{4.145}$$

where $\Phi(t, \tau)$ is the fundamental matrix (state transition matrix) of the homogeneous system

$$\dot{x}_t = F(t)x_t\,,$$

satisfying the matrix differential equation

$$\dot{\Phi} = F(t)\Phi, \qquad \Phi(t, t) = I. \tag{4.146}$$

Taking the expectation in (4.145), we get

$$\hat{x}_t = \Phi(t, t_0)\hat{x}_{t_0}\,,$$

so that

$$\dot{\hat{x}}_t = F(t)\hat{x}_t\,, \tag{4.147}$$

which is the first equation in (4.138).

Now

$$\frac{dP_t}{dt} = \frac{d}{dt}\mathscr{E}\{(x_t - \hat{x}_t)(x_t - \hat{x}_t)^{\mathrm{T}}\} = \mathscr{E}\left\{\left(\frac{d}{dt}(x_t - \hat{x}_t)\right)(x_t - \hat{x}_t)^{\mathrm{T}}\right\}$$

$$+ \mathscr{E}\left\{(x_t - \hat{x}_t)\frac{d}{dt}(x_t - \hat{x}_t)^{\mathrm{T}}\right\}. \tag{4.148}$$

From (4.144) and (4.147), we have

$$\frac{d}{dt}(x_t - \hat{x}_t) = F(x_t - \hat{x}_t) + Gw_t\,,$$

so that

$$dP_t/dt = FP_t + \mathscr{E}\{Gw_t(x_t - \hat{x}_t)^{\mathrm{T}}\} + P_tF^{\mathrm{T}} + \mathscr{E}\{(x_t - \hat{x}_t)\,w_t{}^{\mathrm{T}}G^{\mathrm{T}}\}. \tag{4.149}$$

Now

$$\mathscr{E}\{(x_t - \hat{x}_t)\,w_t{}^{\mathrm{T}}G^{\mathrm{T}}\} = \mathscr{E}\left\{\left[\Phi(t, t_0)(x_{t_0} - \hat{x}_{t_0}) + \int_{t_0}^{t}\Phi(t, \tau)\,G(\tau)w_\tau\,d\tau\right]w_t{}^{\mathrm{T}}G^{\mathrm{T}}\right\}$$

$$= \int_{t_0}^{t}\Phi(t, \tau)\,G(\tau)\,\delta(\tau - t)\,G^{\mathrm{T}}(t)\,d\tau$$

$$= \tfrac{1}{2}G(t)Q(t)\,G^{\mathrm{T}}(t),$$

and, similarly,

$$\mathscr{E}\{Gw_t(x_t - \hat{x}_t)^{\mathrm{T}}\} = \tfrac{1}{2}G(t)Q(t)\,G^{\mathrm{T}}(t).$$

Equation (4.149) therefore becomes

$$dP_t/dt = FP_t + P_tF^{\mathrm{T}} + GQG^{\mathrm{T}},$$

which is (4.138). ▲

Example 4.18. In Section 3.9, Eqs. (3.128)–(3.132), we developed equations of evolution (difference equations) for $\hat{x}$ and P for a discrete linear stochastic system. Using that development, we obtain here differential equations for $\hat{x}_t$ and P_t for the continuous system (4.126) by a formal limiting operation. We consider the white noise equivalent of (4.126), namely, (4.144). For $\Delta = t_{k+1} - t_k$ small, we write (4.144)

$$x_{t_{k+1}} = x_{t_k} + F(t_k)x_{t_k}\Delta + G(t_k)w_{t_k}\Delta + o(\Delta), \tag{4.150}$$

where, according to Example 3.20,

$$\mathscr{E}\{w_{t_k}w_{t_l}^{\mathrm{T}}\} = [Q(t_k)/\Delta]\,\delta_{kl}\,. \tag{4.151}$$

According to the results in Section 3.9,

$$\hat{x}_{t_{k+1}} = [I + F(t_k)\,\Delta]\hat{x}_{t_k},$$

$$P_{t_{k+1}} = [I + F(t_k)\,\Delta]\,P_{t_k}[I + F(t_k)\,\Delta]^{\mathrm{T}} + \Delta^2 G(t_k)[Q(t_k)/\Delta]\,G^{\mathrm{T}}(t_k).$$

These equations are rewritten as

$$(\hat{x}_{t_{k+1}} - \hat{x}_{t_k})/\Delta = F(t_k)\hat{x}_{t_k}, \tag{4.152}$$

$$(P_{t_{k+1}} - P_{t_k})/\Delta = F(t_k)P_{t_k} + P_{t_k}F^{\mathrm{T}}(t_k) + \Delta F(t_k)\,P_{t_k}F^{\mathrm{T}}(t_k) + G(t_k)\,Q(t_k)\,G^{\mathrm{T}}(t_k), \tag{4.153}$$

and taking the $\Delta = 0$ limit we get (4.138). ▲

Consider the scalar *nonlinear* Itô stochastic differential equation,

$$dx_t = f(x_t, t)\,dt + g(x_t, t)\,d\beta_t, \qquad t \geqslant t_0, \tag{4.154}$$

where $\{\beta_t, t \geqslant t_0\}$ has variance parameter $q(t)$. The density $p(x_{t_0})$ is given and x_{t_0} is independent of $\{\beta_t\}$. Since (4.154) is nonlinear, $\hat{x}_t$(4.128) and P_t (4.129) *do not*, in general, characterize $p(x, t)$. They do, however, determine the mean path of (4.154) and the dispersion about that path, and are therefore of great interest. In the next two examples, we develop equations of evolution for $\hat{x}_t$ and P_t by two different methods. In fact, we first develop the equation of evolution for $\mathscr{E}\{\varphi(x_t)\}$, where $\varphi(x_t)$ is an arbitrary function of x_t. Thus we can write down the equation of evolution for any moment of $p(x, t)$. We shall see that these equations are integral-differential equations. The methods we use have apparently been used by others. See Bogdanoff and Kozin [3] and Cummings [5]. Our analysis readily generalizes to the vector case.

Example 4.19. Let $\varphi(x)$ satisfy the hypotheses of Lemma 4.2. Then, by that lemma,

$$d\varphi(x_t) = \varphi_x(x_t)\,dx_t + \tfrac{1}{2}q(t)\,g^2(x_t, t)\,\varphi_{xx}(x_t)\,dt,$$

and substituting for dx_t from (4.154),

$$d\varphi(x_t) = [\varphi_x(x_t)f(x_t, t) + \tfrac{1}{2}q(t)\,g^2(x_t, t)\,\varphi_{xx}(x_t)]\,dt + \varphi_x(x_t)\,g(x_t, t)\,d\beta_t. \tag{4.155}$$

Since

$$\begin{aligned}\mathscr{E}\{\varphi_x(x_t)\, g(x_t, t)\, d\beta_t\} &= \mathscr{E}\{\mathscr{E}\{\varphi_x(x_t)\, g(x_t, t)\, d\beta_t \mid x_t\}\} \\ &= \mathscr{E}\{\varphi_x(x_t)\, g(x_t, t)\, \mathscr{E}\{d\beta_t\}\} \\ &= 0,\end{aligned}$$

upon taking the expectation in (4.155), we get

$$d\mathscr{E}\{\varphi(x_t)\} = \mathscr{E}\{\varphi_x(x_t) f(x_t, t)\}\, dt + \tfrac{1}{2} q(t)\, \mathscr{E}\{g^2(x_t, t)\, \varphi_{xx}(x_t)\}\, dt. \quad (4.156)$$

Now let $\varphi = x_t$. Then we immediately get

$$d\hat{x}_t/dt = \hat{f}(x_t, t), \quad (4.157)$$

where

$$\widehat{(\cdot)} \triangleq \mathscr{E}\{(\cdot)\}.$$

Now letting $\varphi = x_t^2$,

$$d\mathscr{E}\{x_t^2\} = 2\widehat{x_t f}(x_t, t)\, dt + q(t)\, \widehat{g^2}(x_t, t)\, dt. \quad (4.158)$$

But, using (4.157),

$$d(\hat{x}_t^2) = 2\hat{x}_t \hat{f}(x_t, t)\, dt,$$

and combining this with (4.158),

$$\begin{aligned}dP_t/dt &= d\mathscr{E}\{x_t^2\}/dt - d(\hat{x}_t^2)/dt \\ &= 2[\widehat{x_t f}(x_t, t) - \hat{x}_t \hat{f}(x_t, t)] + q(t)\, \widehat{g^2}(x_t, t).\end{aligned}$$

To summarize, $\hat{x}_t$ and P_t satisfy

$$\begin{aligned}d\hat{x}_t/dt &= \hat{f}(x_t, t), \\ dP_t/dt &= 2[\widehat{x_t f}(x_t, t) - \hat{x}_t \hat{f}(x_t, t)] + q(t)\, \widehat{g^2}(x_t, t).\end{aligned} \quad (4.159)$$

Equations (4.159) are not ordinary differential equations. To emphasize this, we write

$$d\hat{x}_t/dt = \int f(x, t)\, p(x, t)\, dx,$$

which in general depends on all the moments of x_t. We note that, if

(4.154) is linear, $f = F(t)x_t$, and g a function of t alone, then (4.159) reduce to

$$d\hat{x}_t/dt = F(t)\hat{x}_t,$$
$$dP_t/dt = 2F(t)[\mathscr{E}\{x_t^2\} - \hat{x}_t^2] + q(t)g^2(t),$$

which is just the result in the linear case (4.138). ▲

Example 4.20. We now derive (4.159) using Kolmogorov's forward equation (4.119). Actually, we derive (4.156), and then (4.159) follow as in Example 4.19.

By definition,

$$\mathscr{E}\{\varphi(x_t)\} = \int \varphi(x)\, p(x, t)\, dx,$$

and therefore

$$d\mathscr{E}\{\varphi(x_t)\}/dt = \int \varphi(x) \frac{\partial p}{\partial t}\, dx. \tag{4.160}$$

Using Kolmogorov's forward equation, (4.160) becomes

$$d\mathscr{E}\{\varphi(x_t)\}/dt = -\int \varphi(x) \frac{\partial(pf)}{\partial x}\, dx + \tfrac{1}{2}q(t) \int \varphi(x) \frac{\partial^2(pg^2)}{\partial x^2}\, dx. \tag{4.161}$$

Evaluating these integrals by parts, and assuming p is well-behaved at infinity, we get

$$d\mathscr{E}\{\varphi(x_t)\}/dt = \int \varphi_x(x) f(x, t)\, p(x, t)\, dx + \tfrac{1}{2}q(t) \int \varphi_{xx}(x)\, g^2(x, t)\, p(x, t)\, dx,$$

which is just (4.156). ▲

Just as we did in Example 4.18 in the linear case, we may derive (4.159) by a limiting operation from the analogous result for the discrete nonlinear system that we derived in Section 3.9. The details of the development are left for the reader.

Example 4.21. If (4.154) is interpreted in the Stratonovich sense, then

the analysis in Example 4.20, but using (4.124) instead of (4.119) as Kolmogorov's equation, leads to

$$d\mathscr{E}\{\varphi(x_t)\}/dt = \mathscr{E}\{\varphi_x f\} + \tfrac{1}{2}q[\mathscr{E}\{\varphi_{xx}g^2\} + \mathscr{E}\{\varphi_x g g_x\}]. \tag{4.162}$$

Then, instead of (4.159), we get

$$\begin{aligned} d\hat{x}_t/dt &= \hat{f} + \tfrac{1}{2}q(t)\,\widehat{gg_x}\,, \\ dP_t/dt &= 2(\widehat{x_t f} - \hat{x}_t\hat{f}) + q(t)(\widehat{g^2} + [x_t g g_x]\hat{}). \quad \blacktriangle \end{aligned} \tag{4.163}$$

As was already noted in Example 4.19, the equations we derived for $\hat{x}_t$ and P_t (4.159) require the knowledge of $p(x, t)$ for their evaluation. We may obtain *approximate* ordinary differential equations for $\hat{x}_t$ and P_t by making certain assumptions about $p(x, t)$. As in Section 3.9, we assume that $p(x, t)$ is symmetric and "close to the mean" (P_t small). Thus, we might neglect third- and higher-order central moments of $p(x, t)$ in evaluating the expectations on the right-hand sides of Eqs. (4.159). The right-hand sides of (4.159) will then depend only on $\hat{x}_t$ and P_t, and these equations can then be solved. As in Section 3.9, we expand in Taylor's series the functions on the right-hand sides of (4.159) and then take their expectation. Using our assumptions about $p(x, t)$, we get the following approximate equations for $\hat{x}_t$ and P_t:

$$\begin{aligned} d\hat{x}_t/dt &= f(\hat{x}_t\,, t) + (P_t/2)f_{xx}(\hat{x}_t\,, t), \\ dP_t/dt &= 2P_t f_x(\hat{x}_t\,, t) \\ &\quad + q(t)[g^2(\hat{x}_t\,, t) + P_t g_x{}^2(\hat{x}_t\,, t) + P_t g(\hat{x}_t\,, t)\, g_{xx}(\hat{x}_t\,, t)]. \end{aligned} \tag{4.164}$$

The comments in footnote 17 of Section 3.9 and the comments at the end of that section with respect to our approximation apply here as well. We also note that equations of evolution for higher-order moments can easily be obtained from (4.156). These higher-order moments can then be retained in the approximate equations.

10. DISCUSSION

In this book we shall use Itô's stochastic integral, and all stochastic differential equations (except for isolated examples) will be understood in the Itô sense. Our reasons for using the Itô integral instead of the Stratonovich integral are the following.

It is much easier to compute expectations of the Itô integral than the Stratonovich integral. Theorem 4.1 does not hold for the Stratonovich

integral. The Itô integral (see Doob [6, Chapter IX]) has other nice mathematical properties not possessed by the Stratonovich integral. As a result, most theoretical work is more conveniently done in the Itô framework. In fact, stochastic stability theory (see Kushner [14]) owes its existence to these properties of the Itô integral. Virtually all the theoretical work in stability and control (see Kushner [13]) for stochastic systems is done using the Itô integral, which provides us with another important reason for using the Itô integral here. The background provided the readers of this book will make this theoretical work more accessible for them. Most important of all is the fact that the Itô integral is defined for a much broader class of functions than the Stratonovich integral. The Stratonovich integral is so restrictive that *it is not applicable to nonlinear filtering theory* (see footnote 3 in Chapter 6).

One might argue that the Stratonovich integral should be used because it is simpler, since it can be manipulated by the formal rules. This is intuitively appealing but beside the point. The Stratonovich integral does not offer any new mathematical insight or content. As a matter of fact, all the results concerning the Stratonovich integral are proved using Itô's theory. Of itself, the Stratonovich integral does not offer any additional physical insight. It does, under certain conditions, more directly model a physical process (Section 8).

We have given (in Sections 6 and 9) transformations from Itô to Stratonovich. For every Itô equation, there is an equivalent Stratonovich equation, and conversely. Thus one and the same process can be modeled via either equation. The transformations we have given enable the interested reader to translate our results to the Stratonovich framework.

REFERENCES

1. K. J. Astrom, On a First Order Stochastic Differential Equation, *Intern. J. Control* 1, 301–326 (1965).
2. A. T. Bharucha-Reid, "Elements of the Theory of Markov Processes and Their Applications." McGraw-Hill, New York, 1960.
3. J. L. Bogdanoff and F. Kozin, Moments of the Output of Linear Random Systems, *J. Acoust. Soc. Am.* **34**, 1063–1066 (1962).
4. J. M. C. Clark, The Representation of Nonlinear Stochastic Systems with Applications to Filtering. Ph. D. Thesis, Electrical Engr. Dept., Imperial College, London, England, 1966.
5. I. G. Cummings, Derivation of the Moments of a Continuous Stochastic System, *Intern. J. Control* **5**, 85–90 (1967).
6. J. L. Doob, "Stochastic Processes." Wiley, New York, 1953.
7. E. B. Dynkin, "Markov Processes." Academic Press, New York, 1965.
8. A. H. Gray, Jr. and T. K. Caughey, A Controversy in Problems Involving Random Parameter Excitation, *J. Math. and Phys.* **44**, 288–296 (1965).

9. K. Itô, Stochastic Integral, *Proc. Imp. Acad. Tokyo* **20**, 519–524 (1944).
10. K. Itô, On Stochastic Differential Equations, *Mem. Amer. Math. Soc.* **4** (1951).
11. K. Itô, Lectures on Stochastic Processes (mimeographed notes), Tata Inst. Fundamental Research, Bombay, India, 1961.
12. N. K. Kulman, A Note on the Differential Equations of Conditional Probability Density Functions. *J. Math. Anal. Appl.* **14**, 301–308 (1966).
13. H. J. Kushner, On the Status of Optimal Control and Stability for Stochastic Systems, *IEEE Intern. Conv. Record, Part* 6 **14**, 143–151 (1966).
14. H. J. Kushner, "Stochastic Stability and Control." Academic Press, New York, 1967.
15. A. V. Skorokhod, "Studies in the Theory of Random Processes." Addison-Wesley, Reading, Massachusetts, 1965.
16. R. L. Stratonovich, "Topics in the Theory of Random Noise." Gordon and Breach, New York, 1963.
17. R. L. Stratonovich, A New Form of Representing Stochastic Integrals and Equations, *J. SIAM Control* **4**, 362–371 (1966).
18. N. Wiener, Generalized Harmonic Analysis, *Acta Math.* **55**, 117–258 (1930).
19. E. Wong and M. Zakai, On the Relation Between Ordinary and Stochastic Differential Equations, *Internat. J. Engrg. Sci.* **3**, 213–229 (1965).
20. E. Wong and M. Zakai, On the Convergence of Ordinary Integrals to Stochastic Integrals, *Ann. Math. Stat.* **36**, 1560–1564 (1965).
21. E. Wong and M. Zakai, On Convergence of the Solutions of Differential Equations Involving Brownian Motion. Rept. No. 65-5, AF-AFOSR-139-64, Electronics Research Lab, Univ. of California, Berkeley, 1965.